

# Weather, extreme heat, and everyday spending in cities: Evidence from longitudinal and nationwide credit card transactions in South Korea

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## ABSTRACT

As weather extremes become more frequent and persistent, cities need to understand how weather conditions are associated with everyday spending. Yet, existing research rarely provides a comprehensive picture of these relationships across spending amount, transaction frequency, sectoral composition, and cross-city variation. By linking nationwide credit card transactions from South Korea covering 2019 to 2020 with city-level meteorological data, this study examines associations of apparent temperature, precipitation, and extreme-heat persistence with everyday spending. Using 20–25 °C as the reference range, the results show that days with apparent temperatures of 33 °C or higher were associated with an estimated 3.44% increase in overall transactions and an estimated 2.30% decrease in overall expenditure. Sectoral patterns were uneven, with higher transactions and expenditure in several retail, restaurant, and recreation categories but lower transactions and expenditure in Home and Internet Shopping. Beyond heat on a given day, a 10-mm increase in precipitation was associated with 1.05% fewer transactions and 0.94% lower expenditure. Meanwhile, each additional extreme-heat day during the preceding three days was associated with declines of 0.34% in transaction frequency and 1.60% in expenditure. A cross-city comparison further suggests that climatic context, coastal location, age structure, and population density were associated with heterogeneity in spending responses to weather. As global warming increases heat exposure, sectors and cities may face different economic gains and losses. Heat-adaptation strategies should therefore account for daily service demand, vulnerable populations, and local weather sensitivity.

## 1. Introduction

Weather extremes are intensifying under anthropogenic warming, with particularly rapid increases in extreme heat. Global assessments show that hot extremes and heatwaves have already increased in frequency, duration, and severity and are projected to intensify further (IPCC, 2023; Meehl & Tebaldi, 2004). These changes are spatially uneven and have been accompanied by accelerating regional trends and more frequent record-breaking events (Fischer et al., 2025; Fischer & Schär, 2010; Perkins-Kirkpatrick & Lewis, 2020). Cities are especially exposed because dense development, impervious surfaces, and altered surface energy balances can amplify local thermal stress (IPCC, 2022; Manoli et al., 2019; Oke, 1982; Zhao et al., 2014).

A broad interdisciplinary literature shows that temperature extremes affect human well-being and economic systems through multiple pathways (Abunyewah et al., 2025). Temperature variation has been

linked to mortality, morbidity, hospitalization, sleep loss, time allocation, labor supply, and productivity across diverse populations and climates (Bigler & Janzen, 2024; Burke et al., 2018; Gasparrini et al., 2015; Graff Zivin & Neidell, 2014; Liao et al., 2025). Economic responses are often nonlinear, with adverse outcomes emerging as temperatures move away from moderate ranges (Burke et al., 2015; Dell et al., 2012; Hsiang et al., 2017). These findings establish the broad social and economic importance of weather stress, yet offer less insight into how everyday urban activities and consumption are reorganized from day to day.

One pathway connecting weather to urban economic activity is the reorganization of daily activities. Adverse thermal and precipitation conditions can increase travel discomfort, alter activity timing and destination choice, and constrain mobility, although these adjustments

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vary across social groups, transport options, and local contexts (Koetse & Rietveld, 2009; Liang & Wang, 2025; Liu et al., 2026; Yücel & Schwanen, 2025; Zheng et al., 2025). Because many purchases are embedded in out-of-home activities, such adjustments can leave observable traces in transaction frequency, expenditure, and spending composition. Consumer spending therefore provides an indirect indication of how everyday activities are maintained, relocated, deferred, or modified under weather stress.

A growing literature shows that weather affects retail performance and consumer spending through changes in purchase timing, basket size, store choice, and spending composition, rather than through a uniform demand shock (Badorf & Hoberg, 2020; Martínez-de-Albéniz & Belkaid, 2021; Murray et al., 2010; Roth Tran, 2023; Seijas et al., 2026; Starr-McCluer, 2000; Tian et al., 2021; Yoo et al., 2024). Existing studies also suggest that precipitation and extreme temperatures can shift spending toward indoor or temperature-controlled environments (Martínez-de-Albéniz & Belkaid, 2021; Roth Tran, 2023; Yoo et al., 2024). Recent evidence that heatwaves reconfigure consumption across categories and transaction channels further emphasizes the value of examining multiple spending margins jointly (Yu et al., 2026). Evidence from Korea also points to strong heterogeneity. Nationwide data indicate declines in aggregate card spending under heatwaves, cold waves, and heavy rainfall (Chang & Cho, 2025). In contrast, evidence from Seoul shows that extremely high and low temperatures can increase sales in brick-and-mortar stores, which is consistent with consumers shifting toward air-conditioned or heated indoor environments (Yoo et al., 2024). These findings indicate substantial heterogeneity in how spending changes with weather conditions.

Drawing on time geography, we view daily consumption as embedded in activity schedules that are constrained by time, mobility, and access to destinations (Ellegård & Svedin, 2012; Hägerstrand, 1970). Weather can tighten these constraints by reducing thermal comfort and increasing the difficulty of travel (Yücel & Schwanen, 2025). Residents may respond by continuing, canceling, deferring, or relocating activities. They may also shift purchases or services to digital channels or change what and how much they buy. These adjustments can appear in transaction frequency, expenditure amount, and category composition. A place-based adaptation perspective further suggests that the feasibility of such responses depends on local exposure, susceptibility, and adaptive capacity (Smit & Wandel, 2006). These conditions vary with climatic, urban, demographic, and socioeconomic context. Together, these perspectives suggest that weather–spending relationships may be nonlinear, differ between transaction frequency and expenditure, vary across spending categories, and depend on city context.

Three gaps motivate this study. First, many studies rely on a single observed dimension of spending, which can obscure differences between transaction frequency and expenditure amount. Distinguishing these dimensions is important for assessing whether weather is related to purchase activity, purchase value, or both. Second, much of the consumer-spending literature centers on contemporaneous temperature, while precipitation and cumulative heat exposure have received less systematic attention. Precipitation can increase the difficulty and discomfort of short daily trips, whereas repeated heat exposure may generate additional adjustments beyond the conditions experienced on a single day (Gössling et al., 2023; Liang & Wang, 2025). Third, nationwide evidence on geographical heterogeneity remains limited. Local climate, coastal setting, demographic composition, urban density, and economic resources may condition weather-related spending patterns, while studies focused on individual cities or sectors provide limited leverage for assessing how consistently these relationships vary across places and consumption activities.

This study addresses these gaps using credit card transaction records for local residents across 152 South Korean cities, linked with daily city-level weather conditions. The analysis is guided by three research questions. First, how does resident spending vary across the temperature spectrum, and do the patterns for transactions and expenditure differ

under cold and warm conditions? Second, beyond daily temperature conditions, how are precipitation and extreme-heat persistence related to transactions and expenditure, and how do these relationships differ across spending categories? Third, how do weather-related spending patterns vary across cities, and which climatic, geographic, demographic, and socioeconomic attributes covary with this heterogeneity? The study contributes by jointly examining transaction frequency and expenditure, distinguishing current weather conditions from recent heat exposure, comparing responses across spending categories, and interpreting cross-city heterogeneity within a place-based conceptual framework.

The remainder of the paper is organized as follows. Section 2 describes the study area, data sources, key weather variables, and resident spending records. Section 3 presents the empirical framework. Section 4 reports the main findings. Section 5 discusses possible behavioral mechanisms, implications, and limitations of the study, followed by the conclusions in Section 6.

## 2. Data

### 2.1. Study area

This study focuses on South Korea, which occupies the southern part of the Korean Peninsula in East Asia. The country covers approximately 100,364 km<sup>2</sup> and had a population of about 51.8 million in 2019. South Korea's climate is influenced by the East Asian monsoon system and complex topography. These conditions produce pronounced seasonality and significant spatial variation in daily meteorological conditions. The warm season, from May to September, is typically hot and humid, often coinciding with monsoon rainfall. In contrast, the cool season, from October to April, is generally colder and drier due to continental high-pressure systems. Geographic variations, including coastal and inland locations as well as basin and mountainous terrain, further contribute to differences across cities in thermal exposure and hydro-meteorological conditions. This heterogeneity provides a useful context for analyzing how weather-related changes in spending vary across locations.

South Korea comprises 17 first-tier administrative divisions, including one Special City, one Special Self-Governing City, six Metropolitan Cities, eight Provinces, and one Special Self-Governing Province (Jeju). These divisions are further divided into second-tier units, including cities (si), counties (gun), and districts (gu). The spatial units used in this study correspond to the geographic resolution of the credit card transaction data. We constructed a nationwide set of 162 administrative units, including the Special City, the Special Self-Governing City, all six Metropolitan Cities, and 154 second-tier units. These units are collectively referred to as “cities” hereafter. Ten cities were excluded because continuous meteorological observations were unavailable during the study period. Eight of the excluded cities were located in Gyeonggi Province, while the remaining two were inland counties in Gyeongsangbuk Province. Two excluded cities were coastal and eight were inland. The final analytical sample therefore comprised 152 cities. The locations and characteristics of the excluded cities are reported in Table B.1.

### 2.2. Meteorological variables and exposure definitions

**Meteorological observations and city-level aggregation.** Meteorological data were obtained from the open data portal of the Korea Meteorological Administration (KMA). We used both the Automated Synoptic Observing System (ASOS) and the Automatic Weather System (AWS) networks. Hourly and daily observations for 2019 and 2020 were collected to construct weather panels at the city-day level. The 2019 panel supports the main analysis, whereas the 2020 observations are used in the extended-window robustness analysis. ASOS and AWS stations were assigned to cities according to their administrative affiliation. We

then extracted observations on air temperature, wind speed, relative humidity, and precipitation at the station level. Hourly observations were used for air temperature, wind speed, and relative humidity, whereas daily totals were used for precipitation.

When multiple stations reported hourly data for the same city, air temperature, wind speed, and relative humidity were averaged across available stations for each city and hour. Hourly air temperature was screened against a plausible range, and outliers were treated as missing values. Missing hourly values in air temperature, wind speed, and relative humidity were interpolated using observations within a  $\pm 3$ -day window centered on the target day, thereby reducing discontinuities at day boundaries. Relative humidity was interpolated using the shape-preserving piecewise cubic Hermite interpolating polynomial (PCHIP) (Fritsch & Butland, 1984). The interpolated values were constrained to the range  $[0, 100]\%$ . Wind speed was interpolated on the  $\log(1+x)$  scale, back-transformed, and constrained to be non-negative. Air temperature was interpolated using cubic interpolation to preserve smooth intra-day variation.

**Apparent temperature and daily exposure measures.** To approximate perceived thermal conditions, this study uses apparent temperature rather than air temperature alone. Apparent temperature combines dry-bulb temperature with other meteorological factors to represent how hot or cold conditions feel to the human body (Steadman, 1984). Hourly apparent temperature was computed from the processed hourly data following the KMA convention (Korea Meteorological Administration (KMA)). Under the KMA definition, apparent temperature in the warm season (May to September) is based on air temperature and humidity. In the cool season (October to April), it follows a wind chill formulation under sufficiently cold and windy conditions. Otherwise, apparent temperature is set equal to air temperature.

We then aggregated the processed hourly data to obtain daily measures at the city level. The primary heat exposure variable is daily maximum apparent temperature, denoted  $AT_{\max}$ . It is defined as the maximum hourly apparent temperature within a day. We also computed the daily mean wind speed from hourly observations to serve as a control variable. Daily precipitation was constructed separately from KMA daily observations. For cities with multiple affiliated stations, we used the largest station-level daily total. This measure captures whether substantial rainfall occurred within a city on a given day and reduces the likelihood that localized rainfall is missed when station coverage is uneven. The final weather panel includes  $AT_{\max}$ , daily precipitation, and daily mean wind speed as the key meteorological variables used in the models.

**Temperature bands.** To model nonlinear relationships between temperature and spending,  $AT_{\max}$  was divided into the following temperature bands:

$$B = \{< 5, [5, 10), [10, 15), [15, 20), [20, 25), [25, 28), [28, 30), [30, 33), \geq 33\}^{\circ}\text{C}. \quad (1)$$

The bands use broader intervals under cooler conditions and finer intervals in the warm tail to distinguish moderately warm, hot, and extreme-heat conditions while retaining sufficient observations within each band. Details of temperature-band coverage across cities are shown in Table B.3. Throughout the analysis, the 20–25°C band is used as the reference band. The 30–33°C band is referred to as the heat band, and  $AT_{\max} \geq 33^{\circ}\text{C}$  is defined as the extreme-heat band. The 33°C cutoff corresponds to the apparent-temperature threshold used in the KMA heatwave warning system. In this study, it classifies individual extreme-heat days.

**Extreme-heat persistence.** To capture cumulative exposure beyond daily temperature conditions, we constructed extreme-heat persistence as the number of extreme-heat days in the preceding three days, excluding the current day:

$$EHP_{ct} = \sum_{k=1}^3 \mathbb{I}(AT_{\max,ct-k} \geq 33^{\circ}\text{C}), \quad EHP_{ct} \in \{0, 1, 2, 3\}, \quad (2)$$

where  $c$  denotes a city and  $t$  denotes a day.

Finally, precipitation enters the regressions in scaled units, with  $P_{ct} = \text{DailyPrecip}_{ct}/10$ . Thus, an increase of one unit in  $P_{ct}$  corresponds to an increase of 10 mm in daily precipitation.

### 2.3. Credit card transactions and spending categories

We use credit card transaction records provided by BC Card, one of the largest card issuers in South Korea. The dataset covers 22 consecutive months, from January 1, 2019, to November 1, 2020. The records are anonymized and aggregated by transaction date, transaction city, cardholder registered city, and detailed merchant category. The main analysis uses observations from 2019, while the 2020 records are used in the extended-window robustness analysis. The analysis focuses on resident spending, defined as transactions in which the cardholder's registered city matches the transaction city. This helps focus the analysis on local everyday consumption while excluding nonlocal purchases and tourist spending. Within the study period and area, the resident dataset comprises more than 5.5 billion transactions and over KRW 140 trillion in expenditure. We focus on two observed dimensions of spending. Transactions capture the frequency of purchase events, whereas expenditure captures total spending amounts. These two dimensions reflect distinct behavioral margins and may change differently under the same weather conditions.

The dataset records consumer spending in more than 200 detailed categories, which are grouped into 34 major categories under BC Card's classification system. The analysis covers overall spending, defined as the sum across all categories, and spending within nine focal categories constructed by reclassifying the detailed categories (Table 1). Overall spending provides an aggregate benchmark for how spending changes with weather conditions across the consumption basket. The nine focal categories were selected to represent essential purchases, service activities, and recreation categories that differ in heat sensitivity and opportunities for indoor or online substitution. For instance, Food Retail includes specialty food outlets such as bakeries and butcher shops. Home and Internet Shopping includes online malls, home-shopping merchants, and payment gateways. Because transaction cities recorded in this category reflected merchant or payment-processing locations, these transactions were assigned to cardholders' registered cities and interpreted as online spending by local residents. Recreation spending is further divided into two categories to reflect differences in activity setting. These differences may affect the potential for heat avoidance and activity substitution under different weather conditions. Indoor Recreation includes activities typically conducted in enclosed environments with temperature control, such as movie theaters, bowling clubs, fitness clubs, billiards halls, and aquariums. Mixed-Exposure Recreation includes activities whose indoor or outdoor setting depends on the venue or season, or is otherwise ambiguous, including golf, tennis courts, swimming pools, general leisure complexes, and ticketed leisure activities.

Because the provider reports aggregate records only when positive spending is observed, we reconstructed a complete daily sequence for every combination of city and category. When no record was present for a given city, category, and day, transactions and expenditure were coded as zero rather than missing. Finally, overall spending was measured for each city and day, whereas the nine focal categories were measured for each city, category, and day.

## 3. Methodology

### 3.1. First-stage estimation of city-level spending changes

The empirical strategy proceeds in two stages. The first stage estimates city-specific associations between weather and spending separately for transactions and expenditure. These estimates are used to characterize the relationship between temperature, precipitation,

**Table 1**

Shares of the nine focal spending categories in the credit card transaction dataset.

| Category                   | Share of transactions (%) | Share of expenditure (%) |
|----------------------------|---------------------------|--------------------------|
| Convenience Stores         | 15.86                     | 4.01                     |
| Supermarkets               | 9.91                      | 5.80                     |
| Restaurants                | 18.69                     | 15.52                    |
| Food Retail                | 6.57                      | 4.07                     |
| Alcohol and Bars           | 0.81                      | 1.58                     |
| Hospitals and Pharmacies   | 6.74                      | 8.75                     |
| Home and Internet Shopping | 14.76                     | 23.49                    |
| Mixed-Exposure Recreation  | 2.05                      | 0.73                     |
| Indoor Recreation          | 0.74                      | 0.89                     |
| Total                      | 76.14                     | 64.83                    |

extreme-heat persistence, and spending across cities. This framework is applied to overall spending and each of the nine focal spending categories. The resulting coefficient estimates and standard errors are obtained for each city or each city-category pair. They are then used as inputs for the second-stage pooling and heterogeneity analyses (Sections 3.2 and 3.3).

Let  $c$  denote a city and  $t$  denote a day. For overall spending, the following city-level regression is estimated:

$$\ln(1 + Y_{ct}) = \alpha_c + \sum_{b \in B \setminus b_0} \beta_{cb} \mathbb{I}(AT_{\max,ct} \in b) + \gamma_c P_{ct} + \delta_c EHP_{ct} + \mathbf{Z}'_t \theta_c + \phi_c W S_{ct} + \varepsilon_{ct}, \quad (3)$$

where  $Y_{ct}$  denotes either daily transactions or daily expenditure in city  $c$  on day  $t$ . The variable  $\mathbb{I}(AT_{\max,ct} \in b)$  equals one if  $AT_{\max}$  falls within temperature band  $b \in B$ . The reference band is  $b_0 = [20, 25]^\circ\text{C}$ .  $P_{ct}$  represents daily precipitation, and  $EHP_{ct}$  denotes extreme-heat persistence, as defined in Section 2.2.  $\mathbf{Z}_t$  includes calendar controls at the day level, namely fixed effects for day of week, month of year, and a national holiday indicator.  $W S_{ct}$  is daily mean wind speed and is included as an additional weather control. The coefficients  $\beta_{cb}$  capture the difference in log spending in city  $c$  when  $AT_{\max}$  falls in band  $b$  relative to the reference band. These estimates are conditional on the other weather variables and controls. Coefficients  $\gamma_c$  and  $\delta_c$  summarize the estimated city-specific associations with precipitation and extreme-heat persistence, respectively.

For category-specific spending, an analogous regression is estimated for each city  $c$  and category  $k$ :

$$\ln(1 + Y_{ckt}) = \alpha_{ck} + \sum_{b \in B \setminus b_0} \beta_{ckb} \mathbb{I}(AT_{\max,ct} \in b) + \gamma_{ck} P_{ct} + \delta_{ck} EHP_{ct} + \mathbf{Z}'_t \theta_{ck} + \phi_{ck} W S_{ct} + \varepsilon_{ckt}, \quad (4)$$

where  $Y_{ckt}$  denotes daily transactions or daily expenditure in city  $c$ , category  $k$ , on day  $t$ . Temperature exposure and other weather variables vary by city and day. They are therefore shared across categories within the same city on the same day.

Eqs. (3) and (4) are estimated separately for each city and city-category panel. Accordingly,  $\alpha_c$  and  $\alpha_{ck}$  are panel-specific intercepts. Models are not estimated for city-category panels with no transactions on any day during the analysis period. For all other panels, zero-valued days from the reconstructed daily sequence are retained using  $\ln(1 + Y)$ . Table B.2 reports the corresponding sample counts, and Table B.4 reports the first-stage model diagnostics.

To assess sensitivity to functional form, we also estimate an alternative specification using a natural cubic spline in  $AT_{\max}$ . This specification produces smooth curves describing the relationship between temperature and spending. Internal knots are placed at the 15th, 50th, and 85th percentiles of the national distribution of city-day  $AT_{\max}$  in 2019. The spline specifications use the same samples and variables, including the  $\ln(1 + Y)$  transformation, month fixed effects, day-of-week controls, holiday indicator, and wind-speed control, as the band

specifications, allowing direct comparison of the estimated patterns. All first-stage regressions are estimated by ordinary least squares with heteroskedasticity-robust standard errors (HC1).

To facilitate interpretation and comparability across cities, spending categories, and weather conditions, the estimates are reported as percentage changes relative to the corresponding reference condition. Since the first-stage models use a log transformation, any coefficient estimate  $\hat{\beta}$  can be converted to a percentage change as

$$\% \Delta = \left[ \exp(\hat{\beta}) - 1 \right] \times 100. \quad (5)$$

This transformation is applied consistently to transactions and expenditure, to overall spending and spending by category, and to all weather variables. Throughout the results, we use “response” to refer to the estimated percentage changes obtained from Eq. (5). The reference condition depends on the weather variable being considered. In the band specification, each temperature coefficient is converted to a percentage change relative to the 20–25°C reference band. Changing the reference band alters the numerical contrasts but leaves the fitted relationship between temperature and spending unchanged (Fig. A.3). For precipitation, the reported estimate corresponds to the percentage change associated with a 10 mm increase in daily precipitation. For extreme-heat persistence, the reported estimate corresponds to the percentage change associated with one additional extreme-heat day in the preceding three days, excluding the current day. For the spline specifications, smooth curves are obtained by evaluating the fitted spline over a set of  $AT_{\max}$  values. Each curve is normalized to equal zero at 22.5°C, the midpoint of the reference band. This normalization facilitates comparison with the band specification and is used to construct the pooled response curves reported in Section 4.

### 3.2. Pooling city-level estimates

The second stage pools the first-stage estimates using inverse-variance weighting. This procedure summarizes the cross-city evidence by giving more weight to more precise estimates. Pooling is used to derive two sets of results. The first is pooled estimates for precipitation and extreme-heat persistence. The second is pooled response curves under the band and spline specifications. Pooling is conducted separately for transactions and expenditure, and separately for overall spending and each focal spending category.

Let  $\hat{\beta}_i$  denote a first-stage estimate, where  $i$  indexes the relevant first-stage estimates. For overall spending,  $i$  corresponds to a city. For a given focal category,  $i$  corresponds to a city-specific estimate within that category. Let  $SE(\hat{\beta}_i)$  denote its corresponding standard error. The pooled estimate is calculated as

$$\hat{\beta}^{pool} = \frac{\sum_i w_i \hat{\beta}_i}{\sum_i w_i}, \quad w_i = \frac{1}{SE(\hat{\beta}_i)^2}. \quad (6)$$

Eq. (6) is applied to the coefficients on scaled precipitation ( $P_{ct}$ ) and extreme-heat persistence ( $EHP_{ct}$ ). The pooled estimates are reported as percentage changes using Eq. (5).

For the band specification, pooling is performed separately for each band  $b \in B \setminus b_0$ , yielding a national step curve relative to the reference band. For the spline specification, normalized response curves from each city are evaluated over a set of  $AT_{\max}$  values and then pooled to obtain a national smooth response curve that is directly comparable to the pooled step curve.

### 3.3. Meta-regressions of cross-city heterogeneity

Pooled estimates provide a national benchmark, but estimated associations with heat and precipitation vary across cities. To examine whether this heterogeneity is related to city characteristics, we perform meta-regressions that link first-stage coefficients to city attributes. Each first-stage estimate is treated as an observation with known precision, and the regressions are estimated using inverse-variance weighting.

**Table 2**

City attributes used in the meta-regressions.

| Variable  | Definition                                                        | Source  |
|-----------|-------------------------------------------------------------------|---------|
| Coastal   | Indicator for coastal cities (1 = coastal, 0 = inland).           | Authors |
| ClimTemp  | City-level 1991–2020 annual climatological mean temperature (°C). | KMA     |
| Share65   | Proportion of residents aged 65 and above.                        | KOSIS   |
| lnGDPpc   | Natural log of GDP per capita (million KRW).                      | KOSIS   |
| lnPopDens | Natural log of population density (persons/km <sup>2</sup> ).     | KOSIS   |

For overall spending, the dependent variable is the first-stage coefficient for each city  $c$ . For spending by category, it is the corresponding coefficient for each city-category pair  $(c, k)$ . We examine heterogeneity for four weather terms: the extreme-heat band ( $AT_{\max} \geq 33^\circ\text{C}$ ), the heat band ( $30 \leq AT_{\max} < 33^\circ\text{C}$ ), precipitation ( $P_{ct}$ , in 10-mm units), and extreme-heat persistence ( $EH P_{ct}$ ). Meta-regressions are estimated separately for transactions and expenditure, and separately for overall spending and each focal spending category. This is consistent with the first-stage regression structure.

Let  $\hat{\beta}_i$  denote a first-stage estimate with standard error  $SE(\hat{\beta}_i)$ . The index  $i$  refers to a city for overall spending and to a city-category pair for spending by category. We estimate

$$\hat{\beta}_i = \eta_0 + \mathbf{X}_i' \boldsymbol{\eta} + u_i, \quad (7)$$

by weighted least squares using inverse-variance weights  $w_i = 1/SE(\hat{\beta}_i)^2$ .

This specification gives greater weight to more precise first-stage estimates. The attribute vector  $\mathbf{X}_i$  includes geographic, climatic, demographic, and socioeconomic characteristics (Table 2). ClimTemp is compiled from KMA station-level climatological normals (1991–2020) and aggregated to the city level using the station matching procedure described in Section 2.2. Coastal is compiled by the authors based on coastline adjacency and administrative boundaries. Share65, lnGDPpc, and lnPopDens are obtained from the Korean Statistical Information Service (KOSIS). These attributes are either time invariant or change slowly relative to daily spending. They are therefore suitable for examining whether differences in the estimated weather associations vary systematically with city characteristics.

The attributes were selected to represent complementary dimensions of heat exposure, susceptibility, and adaptive capacity while remaining consistently available across cities. Coastal and ClimTemp capture regional climatic context because long-term climate and coastal tempering may shape acclimatization, cooling practices, and the meaning of a given absolute temperature. lnPopDens serves as a broad proxy for urban concentration and access to nearby services. Share65 represents age-related susceptibility and potential constraints on activity adjustment. lnGDPpc serves as a proxy for local economic resources that may support adaptation. Prior research similarly identifies typical climate, urban form, age, and socioeconomic disparities as conditions shaping activity changes during heatwaves and emphasizes income-related inequality in urban heat exposure and adaptive capacity (Achebak et al., 2026; Yücel & Schwanen, 2025).

The coefficients in Eq. (7) describe how the magnitude and direction of the first-stage estimates vary with city attributes. Because the dependent variable is a first-stage coefficient from a log spending model, the meta-regression coefficients are interpreted on the log scale. Each coefficient represents the change in the estimated weather association for a one-unit increase in the corresponding city attribute. Tables B.5 and B.6 report descriptive statistics, pairwise correlations, and variance inflation factors for the city attributes used in these models.

### 3.4. Robustness checks and supplementary temporal analyses

The main analysis focuses on 2019 to characterize relationships between weather and spending before the COVID-19 pandemic. As a

temporal robustness check, we repeat the two-stage workflow using an extended panel from January 1, 2019, to November 1, 2020. The 2020 models include categorical controls for five pandemic stages following Sun et al. (2024). All weather terms, calendar controls, and estimation procedures otherwise remain unchanged. Stage definitions and results are reported in Appendix C.

We also conduct two complementary analyses of temporal displacement around extreme-heat events. First, the original city-specific models are re-estimated after excluding city-days without extreme heat during the three calendar days immediately before and after each event. This test examines whether spending potentially shifted to adjacent days affects the comparison sample used to estimate the contemporaneous extreme-heat coefficient. Second, for isolated events with complete windows that do not overlap, an event profile model examines whether adjacent days exhibit systematic offsetting changes, conditional on realized temperature on the current day, extreme-heat persistence, precipitation, and wind speed. Full details are reported in Appendix D.

## 4. Results

### 4.1. Extreme-heat days were infrequent nationwide and unevenly distributed across cities

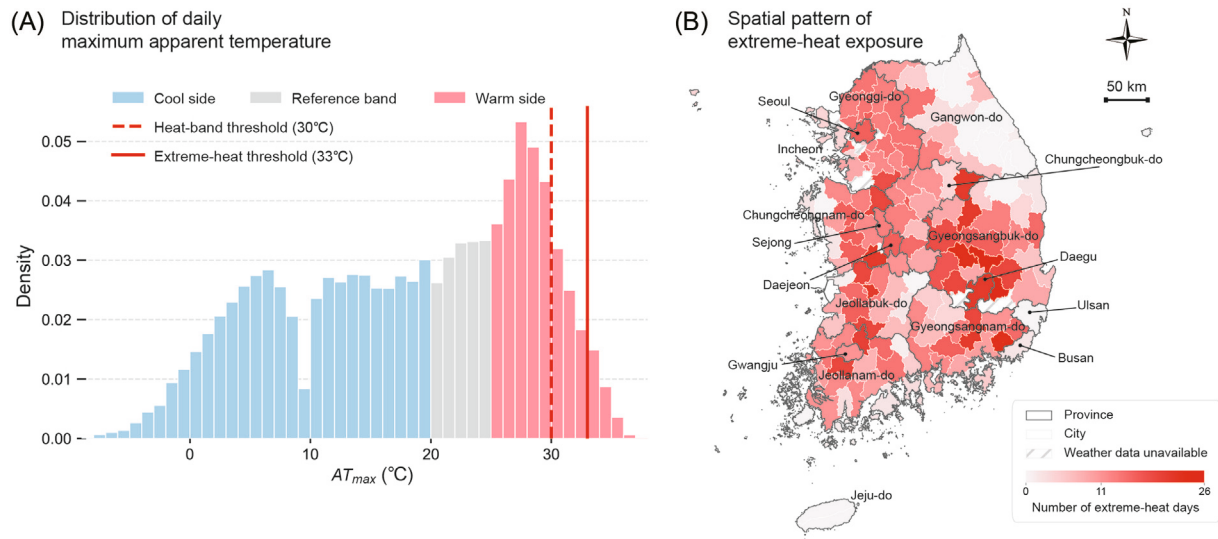
Daily maximum apparent temperature ( $AT_{\max}$ ), defined in Section 2.2, is used to characterize extreme-heat exposure in South Korea. Fig. 1A shows the national distribution of  $AT_{\max}$  in 2019. Most observations were concentrated under relatively mild conditions, while extreme heat appears in the far right tail. The national mean  $AT_{\max}$  was  $18.0^\circ\text{C}$ , and the median was  $19.4^\circ\text{C}$ . Days with  $AT_{\max} \geq 30^\circ\text{C}$  accounted for 10.3% of observations, whereas days in the extreme-heat band ( $AT_{\max} \geq 33^\circ\text{C}$ ) account for 2.8%. This distribution indicates that extreme heat was infrequent at the national level. However, it formed a distinct warm tail where spending patterns may differ from those under more typical warm conditions. The subsequent analysis uses the  $20\text{--}25^\circ\text{C}$  band as the reference band and focuses particularly on the warm tail, including the heat band ( $30\text{--}33^\circ\text{C}$ ) and the extreme-heat band ( $\geq 33^\circ\text{C}$ ).

Fig. 1B shows city-level extreme-heat exposure, measured as the annual number of days with  $AT_{\max} \geq 33^\circ\text{C}$ . Exposure was spatially uneven, showing distinct regional clusters rather than a uniform national pattern. Some cities recorded zero extreme-heat days, mainly in north-eastern mountainous regions and several coastal and island areas. By contrast, higher exposure was concentrated inland, especially around Daegu and southern Gyeongsangbuk-do. Relatively elevated exposure was also observed near the Seoul metropolitan area. Exposure ranged from 0 to 26 extreme-heat days across cities in 2019, highlighting substantial variation across cities in heat exposure.

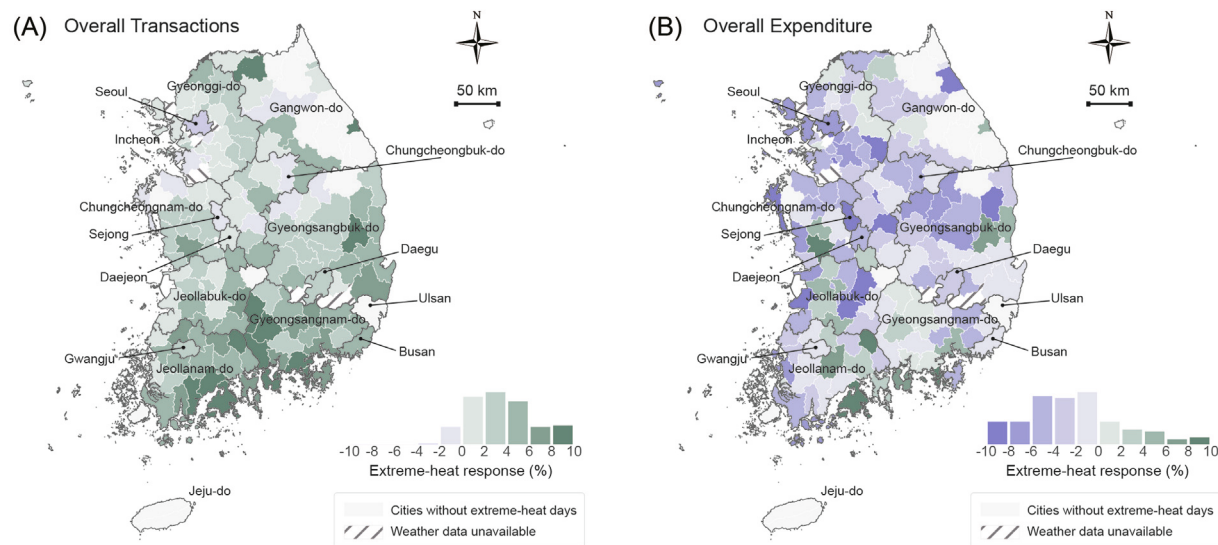
Overall, extreme-heat exposure in 2019 was infrequent at the national level but spatially concentrated in specific inland city clusters. Hatched areas in Fig. 1B indicate cities excluded from the analytical sample because of weather data availability or continuity constraints. Together, these patterns provide descriptive context for the subsequent analysis of nonlinear spending responses and heterogeneity across cities.

### 4.2. Extreme heat was associated with increases in overall transactions and decreases in overall expenditure across most cities

Fig. 2 maps city-level estimates for the extreme-heat band ( $AT_{\max} \geq 33^\circ\text{C}$ ), expressed as percentage changes relative to the reference band. The analysis covers 152 cities, of which 18 recorded no extreme-heat days in 2019 and therefore had no estimable coefficient for this band. Among the remaining 134 cities, 90.3% had positive transaction estimates, and the median estimated change was 3.33%. Positive transaction estimates were geographically widespread, while negative



**Fig. 1.** Daily maximum apparent temperature and extreme heat exposure across cities in South Korea, 2019. (A) Distribution of daily maximum apparent temperature ( $AT_{max}$ ) across all city-day observations. (B) Annual number of extreme-heat days ( $AT_{max} \geq 33^\circ\text{C}$ ) by city. Hatched areas indicate cities with unavailable weather data.



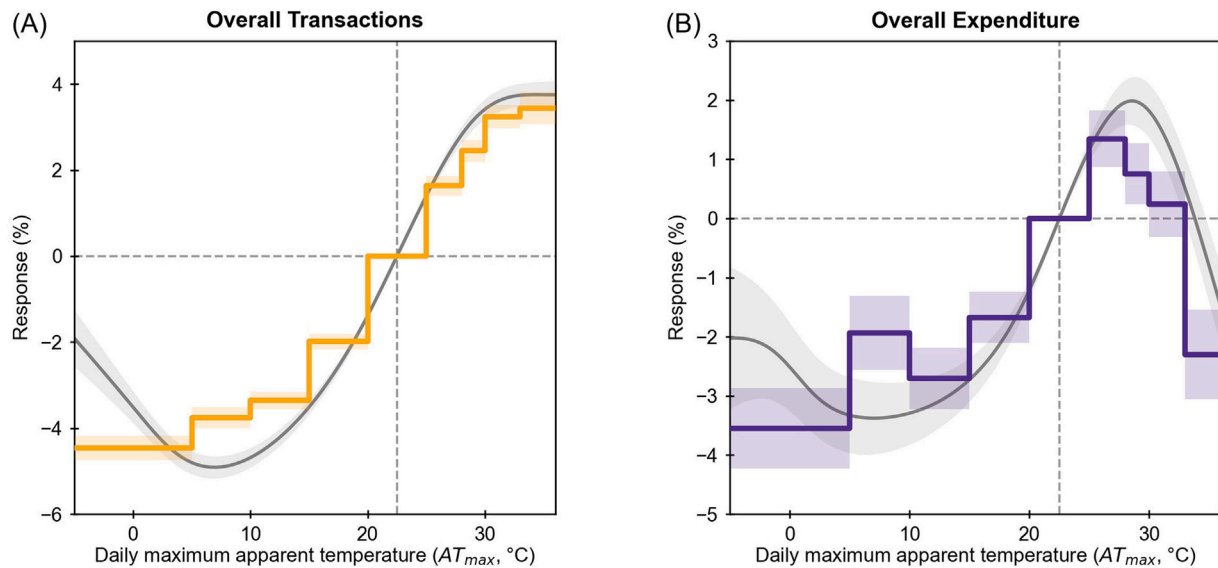
**Fig. 2.** City-level patterns of spending changes associated with extreme heat in South Korea, 2019. Maps report first-stage city-level regression estimates for the extreme-heat band ( $AT_{max} \geq 33^\circ\text{C}$ ), expressed as percentage changes relative to the reference band. (A) Overall transactions. (B) Overall expenditure. Histogram legends summarize the distribution of estimates within each panel.

estimates occurred in a smaller set of cities. Overall expenditure estimates were negative in 74.6% of the 134 cities, with a median estimated change of  $-2.39\%$ . Thus, the city-level estimates more often combined increases in transaction frequency with decreases in expenditure under extreme heat.

Figs. A.1 and A.2 further show that this divergence differed across spending categories rather than reflecting a uniform nationwide pattern. Positive estimates predominated across cities for Convenience Stores, Supermarkets, and Restaurants and were also common for both recreation categories. Negative estimates predominated for Home and Internet Shopping, while Hospitals and Pharmacies showed negative or spatially mixed patterns.

#### 4.3. Temperature showed divergent relationships with transaction frequency and expenditure amount

Fig. 3 shows the national pooled response curves for overall transactions and expenditure in 2019, expressed as percentage changes relative to the reference band. For transactions, the pooled band estimates increased from the cool side toward the warm side. Relative to the reference band, the estimate was  $-4.46\%$  below  $5^\circ\text{C}$  and became progressively less negative as temperature approached the reference band. It was positive above the reference band, reaching  $3.24\%$  in the heat band and  $3.44\%$  in the extreme-heat band. For expenditure, the pooled response curve was lower at both temperature tails and higher



**Fig. 3.** Pooled response curves relating daily maximum apparent temperature ( $AT_{max}$ ) to overall spending in South Korea, 2019, expressed as percentage changes relative to the reference band. The step function shows pooled responses from the band specification, and the smooth curve shows pooled responses from the spline specification. Shaded areas indicate 95% confidence intervals. (A) Overall transactions. (B) Overall expenditure.

immediately above the reference band. The estimate was  $-3.55\%$  below  $5^{\circ}\text{C}$ , rose to  $1.35\%$  in the  $25\text{--}28^{\circ}\text{C}$  band, and then declined to  $0.75\%$  at  $28\text{--}30^{\circ}\text{C}$  and  $0.24\%$  in the heat band. In the extreme-heat band, the pooled expenditure estimate was  $-2.30\%$  relative to the reference band. The negative warm-side estimate was concentrated in the extreme-heat band. Taken together, the extreme-heat band combined a positive transaction estimate with a negative expenditure estimate. Both transaction and expenditure estimates were also negative on the cool side, with a larger negative transaction estimate in the lowest band.

Figs. 4 and 5 show that the pooled response curves varied substantially across categories. For Convenience Stores, Supermarkets, Restaurants, and Alcohol and Bars, transaction estimates were negative on the cool side and became increasingly positive across the warm bands. Their extreme-heat estimates ranged from  $4.47\%$  to  $7.23\%$ . The extreme-heat transaction estimates were also positive for Mixed-Exposure Recreation and Indoor Recreation, at  $4.18\%$  and  $4.87\%$ , respectively. The Food Retail transaction estimates varied comparatively little, whereas the estimates for Hospitals and Pharmacies and Home and Internet Shopping were negative in the extreme-heat band, at  $-2.99\%$  and  $-3.43\%$ , respectively. For expenditure, extreme-heat estimates were positive for Convenience Stores, Supermarkets, Restaurants, Food Retail, and Mixed-Exposure Recreation. The largest were  $9.21\%$  for Supermarkets and  $7.81\%$  for Food Retail, while the Mixed-Exposure Recreation estimate was  $3.59\%$ . These positive category estimates contrasted with the negative overall expenditure estimate, showing that the pooled overall estimate masked substantial variation across spending categories. Home and Internet Shopping was the clearest exception, with expenditure estimates becoming increasingly negative above the reference band and reaching  $-9.91\%$  under extreme heat. Hospitals and Pharmacies also differed across the two spending dimensions. Transaction estimates were negative at the warm tail, whereas the extreme-heat expenditure point estimate was modestly positive but not statistically significant.

#### 4.4. Precipitation and extreme-heat persistence showed different associations with spending

Fig. 6 presents the pooled estimates for precipitation and extreme-heat persistence, conditional on temperature bands and the other controls. Both variables were negatively associated with overall transactions and expenditure, but their category patterns differed.

A  $10\text{ mm}$  increase in daily precipitation was associated with estimated decreases of  $1.05\%$  in overall transactions and  $0.94\%$  in overall expenditure. One additional extreme-heat day in the preceding three days was associated with decreases of  $0.34\%$  and  $1.60\%$ , respectively. The persistence estimate was therefore more negative for expenditure than for transactions after controlling for the current day's temperature band.

At the category level, precipitation estimates were negative for both spending dimensions in Convenience Stores, Supermarkets, Restaurants, Food Retail, Hospitals and Pharmacies, and Mixed-Exposure Recreation. The largest negative expenditure estimate was observed for Mixed-Exposure Recreation ( $-1.95\%$ ), together with a smaller transaction estimate of  $-0.16\%$ . Estimates were positive for both dimensions in Alcohol and Bars, Home and Internet Shopping, and Indoor Recreation. Among these categories, the estimates ranged from  $0.19\%$  to  $0.39\%$  for transactions and from  $0.16\%$  to  $0.39\%$  for expenditure.

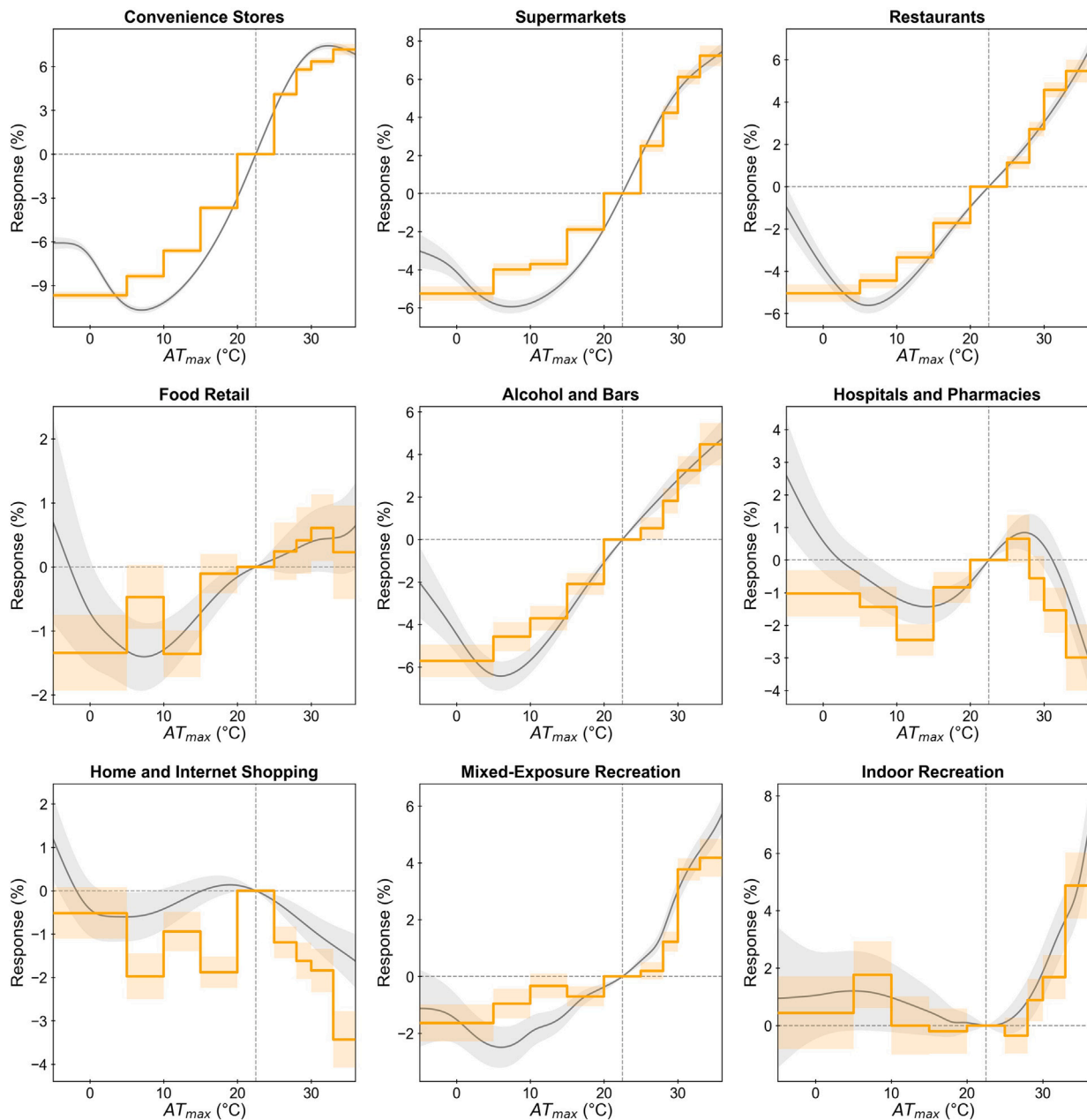
Estimates for extreme-heat persistence showed a different pattern. Estimates were negative for both dimensions in Convenience Stores, Supermarkets, Food Retail, and Home and Internet Shopping. The largest negative expenditure estimate was  $-2.90\%$  for Home and Internet Shopping. Transaction estimates were positive for Restaurants ( $0.97\%$ ), Alcohol and Bars ( $0.74\%$ ), Mixed-Exposure Recreation ( $1.28\%$ ), and Indoor Recreation ( $3.42\%$ ). For expenditure, the positive estimates were statistically significant for Restaurants ( $1.44\%$ ) and Indoor Recreation ( $2.19\%$ ), whereas the estimates for Alcohol and Bars and Mixed-Exposure Recreation were not statistically distinguishable from zero. The persistence estimate for Hospitals and Pharmacies was negative for transactions but close to zero for expenditure.

Overall, precipitation and extreme-heat persistence were associated with different category patterns. Both were negatively associated with overall transactions and overall expenditure, but the positive category estimates were concentrated in different activities and differed between transactions and expenditure.

#### 4.5. City attributes were associated with heterogeneity in spending estimates

Fig. 7 summarizes the exploratory meta-regressions linking city attributes to variation in first-stage estimates. Results are shown separately for transactions and expenditure.

Climatic context produced the broadest recurring pattern. Higher ClimTemp was generally associated with more positive warm-side



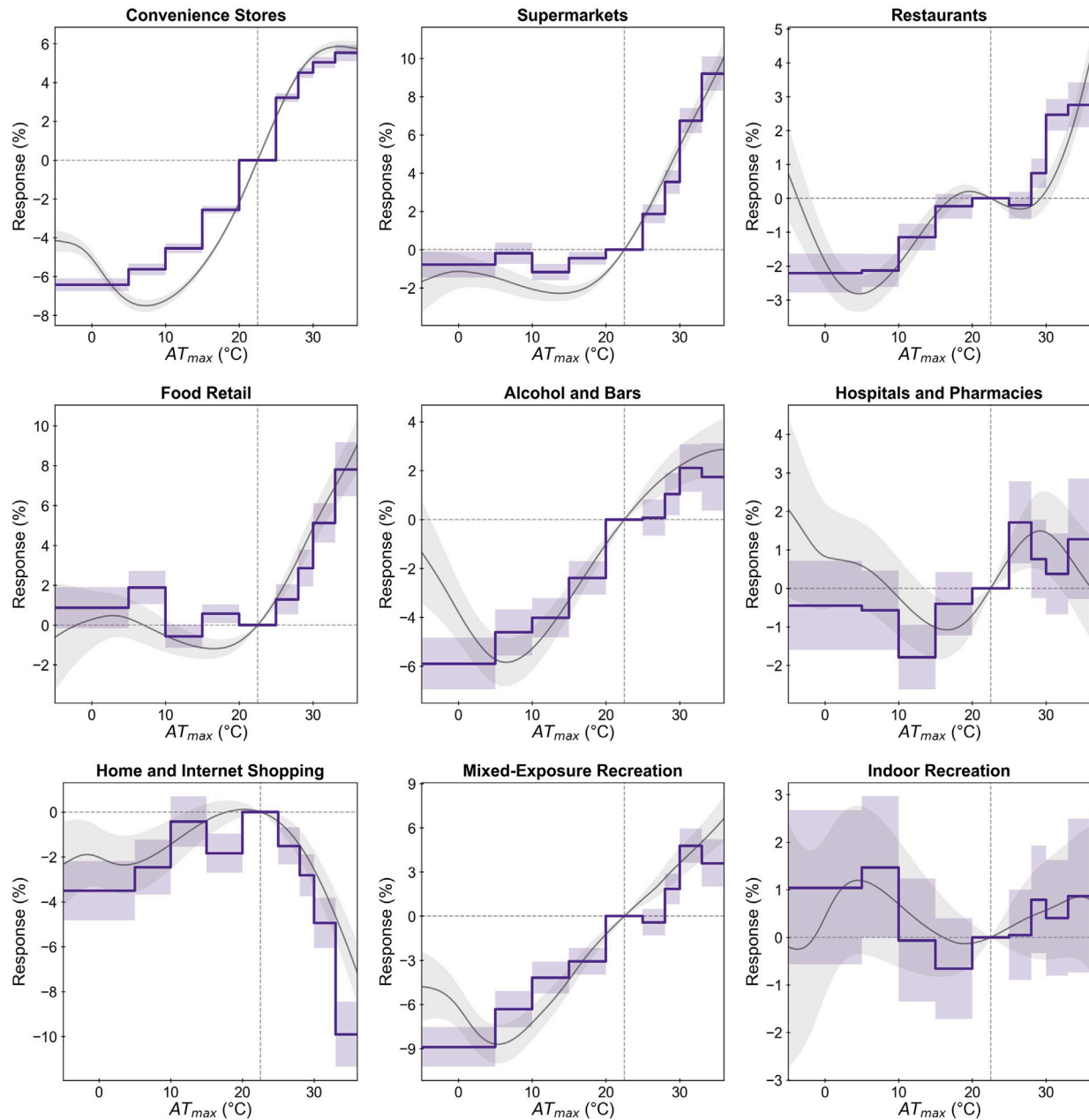
**Fig. 4.** Category-specific pooled response curves relating daily maximum apparent temperature ( $AT_{max}$ ) to transactions in 2019, expressed as percentage changes relative to the reference band. Each panel reports pooled responses for one focal spending category. Step functions represent the band specification, smooth curves represent the spline specification, and shaded areas indicate 95% confidence intervals.

transaction coefficients in overall spending and several routine categories. For the extreme-heat band, the ClimTemp coefficients ranged from 0.005 for Supermarkets to 0.021 for Hospitals and Pharmacies. Home and Internet Shopping differed from this broader pattern, with a coefficient of  $-0.013$ . By contrast, higher ClimTemp was associated with more negative extreme-heat-persistence coefficients for overall transactions and several categories. Its relationships with precipitation estimates were generally smaller. These results indicate that long-term climate was related differently to responses to daily temperature conditions and recent heat exposure.

Age structure was associated mainly with heterogeneity in transaction frequency. Higher Share65 was positively associated with extreme-heat transaction estimates for Supermarkets (0.251), Food Retail (0.224), Hospitals and Pharmacies (0.279), and Home and Internet Shopping (0.148). It was negatively associated with Mixed-Exposure

Recreation ( $-0.268$ ). Higher Share65 was also associated with more negative precipitation estimates for overall transactions and several routine categories, but with positive estimates for Alcohol and Bars and Home and Internet Shopping. For extreme-heat persistence, its transaction coefficients were positive for Home and Internet Shopping and negative for both recreation categories. Corresponding expenditure relationships were less widespread.

Coastal location and population density showed more selective relationships. Coastal cities had more positive extreme-heat transaction coefficients for Supermarkets and Mixed-Exposure Recreation, but a more negative coefficient for Hospitals and Pharmacies. Coastal location was also associated with selected precipitation and persistence estimates, with directions varying across categories. Population density was related mainly to transaction-frequency heterogeneity. Denser cities tended to have weaker positive heat-related transaction estimates



**Fig. 5.** Category-specific pooled response curves relating daily maximum apparent temperature ( $AT_{max}$ ) to expenditure in 2019, expressed as percentage changes relative to the reference band. Each panel reports pooled responses for one focal spending category. Step functions represent the band specification, smooth curves represent the spline specification, and shaded areas indicate 95% confidence intervals.

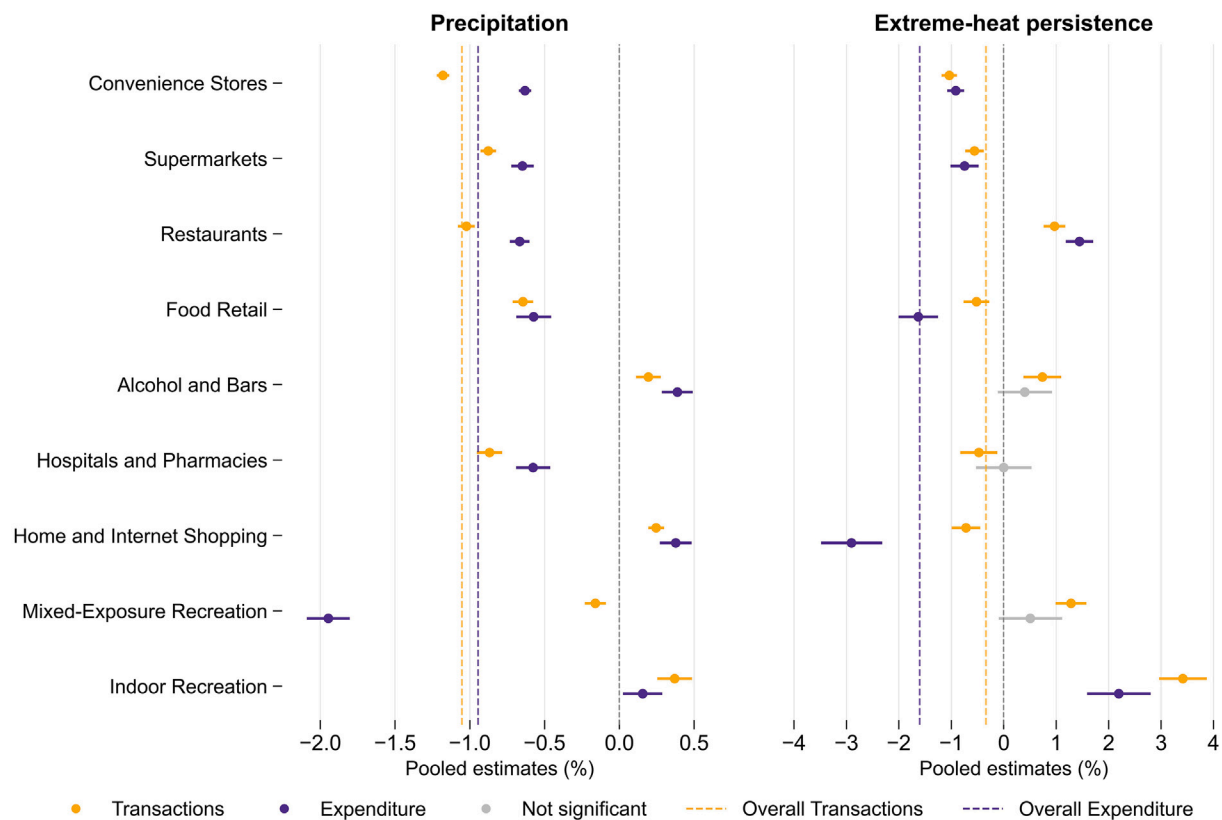
in overall spending and several routine categories, but a more positive extreme-heat estimate for Home and Internet Shopping. Its precipitation coefficients were also negative for overall transactions and several categories.

GDP per capita showed fewer recurring relationships and no common direction across categories. Overall, the meta-regression results point to category-specific geographical conditioning rather than a single ranking of cities by vulnerability or adaptive capacity. The relationships also appeared more consistently for transactions than for expenditure, suggesting that city context was more closely related to heterogeneity in purchase frequency than in total spending amounts.

## 5. Discussion

### 5.1. Key findings and interpretations

This study presents nationwide evidence on how weather conditions, including temperature, precipitation, and extreme-heat persistence, are associated with resident spending across South Korean cities. By distinguishing transactions from expenditure, the analysis shows that weather is associated with multiple observed dimensions of everyday spending rather than with a uniform contraction or expansion. The conceptual framework provides a structure for interpreting these



**Fig. 6.** Pooled estimates for precipitation ( $P_{ct}$ ) and extreme-heat persistence ( $EHP_{ct}$ ) for overall spending and spending categories in 2019. Points represent estimated percentage changes, and horizontal bars indicate 95% confidence intervals.

differences in relation to activity participation, timing, destination, purchase channel, and purchase value.

The main results can be summarized in three findings. First, relationships between temperature and spending are nonlinear, with transactions and expenditure exhibiting distinct response patterns. Transaction estimates became increasingly positive toward the warm tail, whereas expenditure estimates were positive immediately above the reference band but became negative under extreme heat. Second, precipitation and extreme-heat persistence are associated with additional differences in spending beyond daily temperature conditions. Both are linked to lower overall transactions and overall expenditure, but their category-specific patterns differ substantially. Third, weather–spending associations vary across cities and activities, with climatic context, age structure, coastal location, and population density related to selected dimensions of this variation.

Prior evidence links colder conditions to reduced routine mobility, which may help explain the negative cool-side estimates (Koetse & Rietveld, 2009). The different heat-related patterns in transaction frequency and expenditure help reconcile existing Korean evidence. The decline in overall expenditure is consistent with nationwide evidence of lower aggregate card spending during adverse weather (Chang & Cho, 2025). At the same time, the positive expenditure estimates in several in-person retail categories are consistent with evidence from Seoul reporting higher retail sales under extreme temperatures, which the study interprets as thermal comfort seeking (Yoo et al., 2024). These findings need not conflict because overall expenditure can decline even when spending increases in particular categories. This divergence suggests that weather-related spending differences may involve purchase value and category composition as well as purchase frequency, underscoring the importance of examining both spending dimensions.

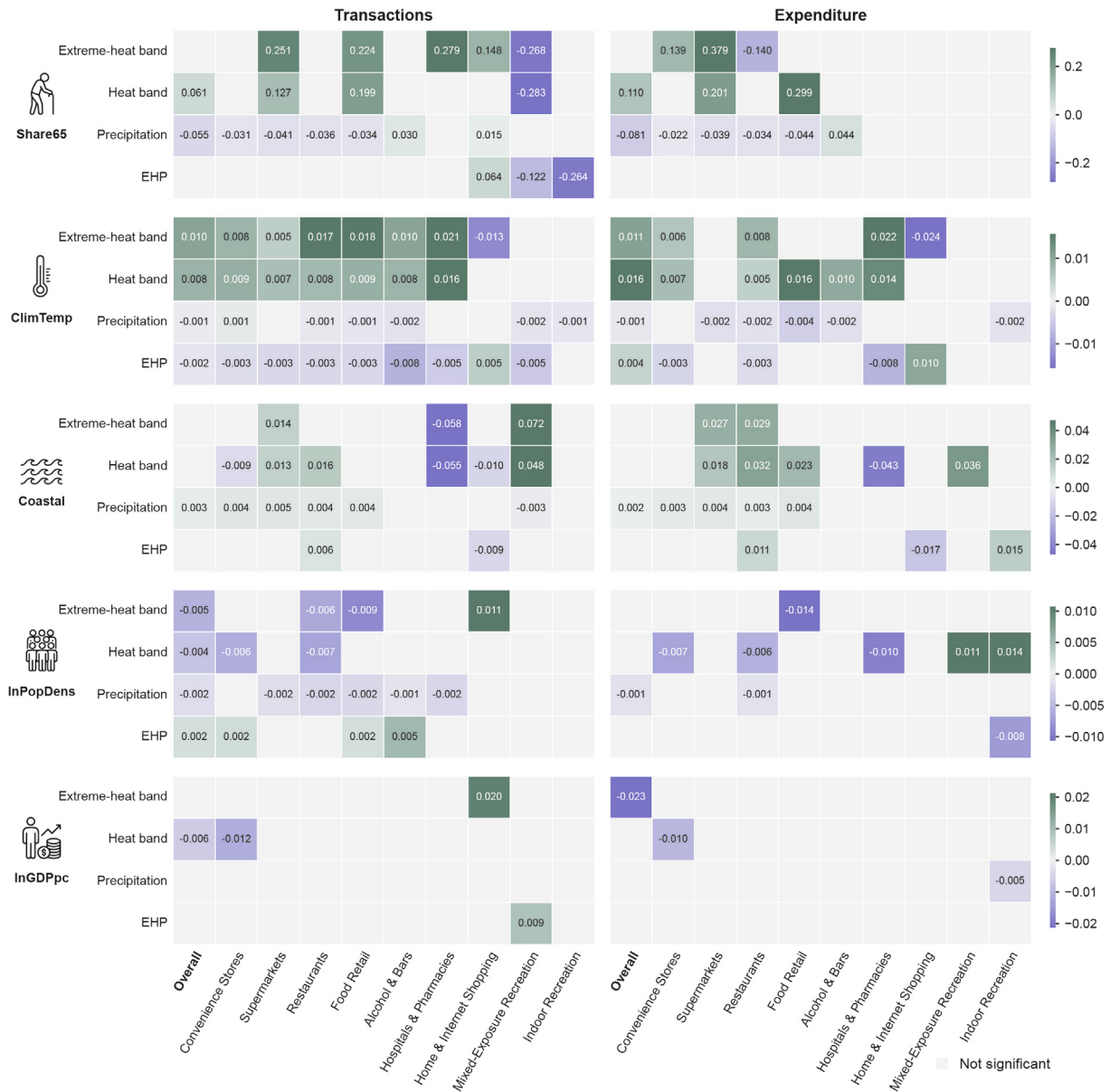
Under extreme heat, expenditure estimates were positive for Convenience Stores, Supermarkets, Restaurants, Food Retail, and Mixed-Exposure Recreation, but negative for Home and Internet Shopping.

This contrast suggests differential weather sensitivity across selected in-person categories and Home and Internet Shopping. Hospitals and Pharmacies had a negative transaction estimate, whereas the expenditure point estimate was modestly positive but not statistically significant. This difference may indicate fewer recorded purchase events without a corresponding decline in expenditure.

In addition to temperature associations, precipitation and extreme-heat persistence capture two distinct forms of weather exposure. Prior mobility research has linked rainfall to fewer short trips and sustained heat to changes in mobility and activity schedules (Gössling et al., 2023; Liang & Wang, 2025). In this study, precipitation and extreme-heat persistence were negatively associated with overall transactions and expenditure, but their category-specific estimates varied. The negative overall associations may reflect greater difficulty completing short trips during rainfall and repeated adjustments to activity schedules during sustained heat. The positive estimates in selected service and recreation categories may suggest that some spending was retained or redirected toward these activities.

The geographical heterogeneity suggests that local climatic, urban, and demographic conditions may shape weather-related spending patterns across South Korean cities. Long-term climate may be related to acclimatization and routine cooling practices, while coastal influence may modify local thermal conditions. Population density may affect the proximity and accessibility of services, while age structure may be associated with heat susceptibility and flexibility in daily schedules. The variation across weather terms and spending categories further suggests that several aspects of local context may be relevant to spending responses.

The extended analysis including 2020 preserved the broad nonlinear and category-specific patterns, although several estimates changed in magnitude, indicating broad stability alongside some sensitivity to pandemic disruptions. The supplementary temporal analyses also showed that removing adjacent days without extreme heat did not alter



**Fig. 7.** Meta-regression coefficients linking city attributes to heterogeneity in first-stage estimates. Each heatmap corresponds to one city attribute. Rows represent the extreme-heat band ( $AT_{\max} \geq 33^{\circ}\text{C}$ ), the heat band ( $30 \leq AT_{\max} < 33^{\circ}\text{C}$ ), precipitation ( $P_{cr}$ ), and extreme-heat persistence ( $EHP_{cr}$ ). Columns represent overall spending and the nine focal spending categories. Cell color indicates coefficient sign and magnitude. Annotations identify coefficients with  $p < 0.05$  in the corresponding meta-regression.

the overall pattern of higher transactions and lower expenditure. The pre- and post-event estimates showed no positive spending accumulation before events and no systematic rebound afterward, providing no evidence of systematic three-day temporal displacement around the analyzed events. Detailed results are reported in [Appendices C and D](#).

Overall, these findings show that weather conditions are associated with differences in purchase frequency, spending amount, and category-specific spending patterns, with substantial variation across cities and activities. These patterns suggest that residents may adjust when, where, and how they consume, while the scope of such adjustments may vary across climatic, urban, and demographic contexts. Rising heat may therefore be associated with uneven gains and losses across spending categories and cities rather than a uniform economic loss.

## 5.2. Policy implications

The findings have several implications for how cities plan heat adaptation, provide public services, and use spending data to assess climate-related impacts. First, weather–spending relationships may be described incompletely when only transaction counts or only expenditure is examined. Transactions capture purchase frequency, whereas expenditure represents the amount spent. Monitoring systems and impact assessments should therefore retain both dimensions. This distinction is important because weather was associated differently with transaction frequency and expenditure at both the overall and category levels.

Second, daily temperature conditions and extreme-heat persistence should be considered separately. Extreme-heat persistence was negatively associated with overall transactions and expenditure, but Restaurants and selected recreation categories showed positive associations.

Spending across daily services therefore did not follow a uniform pattern as hot conditions continued. Cities should maintain access to different types of services and provide adequate cooling options during multi-day heat events. Shaded streets may support street commercial activity under hot conditions (Zhang et al., 2026). Cooling options can also include community centers, senior welfare centers, libraries, and public cultural venues. These measures may be particularly important in neighborhoods with older populations or limited access to private cooling.

Third, prior research suggests that rainfall can make short trips less convenient (Gössling et al., 2023). The negative precipitation associations in this study are therefore consistent with greater difficulty making routine purchases outside the home. Cities can improve sheltered pedestrian routes in commercial districts and provide covered access to public transport stations and bus stops. Reliable transit operations, clear service information, and barrier-free last-mile connections are also important during heavy rainfall. Together, these measures may help residents reach retail and other daily services, especially older adults and people with limited mobility.

Finally, the meta-regression findings show that weather–spending relationships vary across cities. Local climate showed the broadest recurring pattern, while coastal setting, population aging, and population density were related to selected responses. These differences support adapting implementation to local conditions rather than applying the same approach everywhere.

South Korea's existing heat-response system provides a national framework for making these local adjustments. The government has expanded designated heatwave shelters beyond public facilities to bank branches, rail stations, and large retail stores (Ministry of the Interior and Safety (MOIS), 2026). Seoul has also designated selected 24-hour convenience stores as climate shelters to improve access beyond public facilities with limited opening hours (Seoul Metropolitan Government, 2024). This expansion increases the available locations, but not necessarily their accessibility or use. Nationwide evidence identifies socioeconomic differences in travel-time access to cooling centers (Lee & Chae, 2021). A survey in Gwangju also reports low awareness and use, along with concerns about location, opening hours, and information (Oh et al., 2025). Local planning should therefore consider accessibility, operating conditions, and public awareness as well as facility numbers.

Our results add information about how spending patterns vary with weather conditions, extreme-heat persistence, spending category, and city. Local governments could consider this evidence alongside heat-risk and accessibility measures when deciding where routine services need more support. The contrast between daily temperature conditions and extreme-heat persistence suggests that service arrangements may also need to change as hot conditions persist. During multi-day heat episodes, local authorities could extend the hours and staffing of appropriate cooling facilities, strengthen welfare outreach, and provide clearer information about nearby cooling sites and essential services. Accordingly, local governments can leverage insights from consumer spending data to tailor these heat-response measures to the specific needs of different cities.

Although this analysis focuses on South Korea, the framework may be relevant to other regions with similar climatic, demographic, urban, and payment characteristics. It is most relevant to highly urbanized areas with hot and humid summers, pronounced seasonal rainfall, aging populations, and widespread electronic payments. Examples include Japan, parts of eastern and coastal China, and other humid subtropical metropolitan regions where daily consumption depends on short trips and access to indoor services. This broader relevance is consistent with the policy focus on urban heat in East Asia. The World Bank notes that urban heat increasingly threatens the competitiveness, livability, and inclusiveness of East Asian cities. It recommends measures to cool urban spaces, protect vulnerable populations, and integrate heat resilience into city strategies and operations (Roberts et al., 2023). The broader framework of examining both spending dimensions and considering local conditions may therefore support comparative research elsewhere.

### 5.3. Limitations and future work

Several limitations should be noted. First, the two-stage framework is observational and is designed to characterize associations and non-linear response patterns rather than to provide causal estimates. The first-stage regressions use daily variation within each city and incorporate calendar controls, daily weather conditions, and extreme-heat persistence. However, unobserved shocks correlated with weather may still influence the estimates. Month fixed effects account for average differences in spending across calendar months, but residual variation related to vacations, promotions, or other demand shocks may remain. The main analysis also covers only one pre-pandemic year, which limits the assessment of longer-term variation. In addition, extreme-heat persistence is measured as the number of days with  $AT_{\max} \geq 33^{\circ}\text{C}$  during the preceding three days. This indicator does not fully capture the intensity or complete duration of a heat spell.

Second, the spending data include transaction counts and expenditure recorded in the BC Card system. Although the dataset is extensive and nationwide, it excludes cash payments and transactions conducted through other payment networks. The level and composition of observed spending may therefore differ from total household consumption, and the estimated associations should be interpreted as patterns in recorded card spending. Nevertheless, because BC Card is among the largest card issuers and credit card payments are widely used in South Korea, the dataset still provides a useful perspective on broader patterns of residents' consumption behavior.

Third, behavioral adjustments are interpreted from spending patterns rather than directly observed. Changes in activity participation, destination, purchase timing, and transaction channel are plausible interpretations of the differences across spending dimensions and categories, but trips, venue visits, and time use are not observed. The supplementary temporal analyses reduce concern that the main estimates are primarily explained by systematic displacement across adjacent days, but they cannot identify anticipation based on weather forecasts or adjustments within the day. Future research linking spending records with mobility, footfall, or time-use data could examine these adjustments more directly.

Fourth, the spatial and temporal resolution of the data limits the measurement of weather exposure. Spending is observed at the city and daily levels, without information on the specific establishments, neighborhoods, or times at which purchases occurred. Although transactions are assigned to cities, the data therefore cannot identify the local thermal conditions experienced at the actual place and time of consumption. The city-level meteorological data represent regional ambient conditions rather than neighborhood-level or personal exposure and may overlook local heat variation associated with urban form, land cover, and ventilation. The meta-regressions are also limited to attributes available consistently across the cities and do not directly measure urban consumption structure, digital adoption, retail density, or service accessibility. Future work linking more spatially and temporally detailed spending records with finer-scale weather and urban-service data could examine these dimensions more directly.

Finally, external validity should be considered. Because the analysis is based on South Korea, the documented patterns may differ in settings with other climates, urban forms, consumption structures, and payment systems. Replication over longer periods and in other regions would help assess the generality of the findings. Future research could also combine weather-related spending estimates with climate projections to examine potential patterns under future warming.

## 6. Conclusions

This study examines how temperature conditions, precipitation, and extreme-heat persistence were associated with resident spending across South Korean cities. The analysis links nationwide credit card records for local residents with daily weather data for each city and examines

variation across transactions and expenditure, spending categories, and cities.

The results show that the relationship between weather and spending cannot be summarized by a single pattern. Temperature associations were nonlinear, and the response curves for transactions and expenditure differed. Transaction estimates became increasingly positive toward the warm tail, whereas expenditure estimates were positive immediately above the reference band but became negative under extreme heat. The extreme-heat band therefore combined a positive overall transaction estimate with a negative overall expenditure estimate. Precipitation and extreme-heat persistence had additional negative associations with overall spending after daily temperature conditions were controlled, while their category-specific patterns varied. Together, these findings show that adverse weather is associated with uneven spending patterns across categories rather than a uniform gain or loss in urban consumption.

The associations between weather and spending also varied across cities. Climatic context, coastal location, age structure, and population density were related to selected dimensions of this heterogeneity, although their relationships differed across weather terms and spending categories. This geographical variation highlights the importance of local context when interpreting weather-related spending patterns. Taken together, the findings connect estimated differences in purchase frequency and spending amount with variation across activities and city contexts. The implications of rising heat for urban spending are therefore unlikely to follow a uniform pattern across places.

### CRedit authorship contribution statement

**Mengqi Sun:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Yang Xu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization. **Xuanqing Wang:** Writing – review & editing. **Sangwon Park:** Writing – review & editing, Data curation.

### Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to check grammar and improve clarity and readability. The authors also used Google Gemini to generate the icons used in Fig. 7 to help illustrate the meanings of the city attributes. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supplementary figures

#### A.1. Category-specific maps of extreme-heat responses

See Figs. A.1 and A.2.

**Table B.1**

Cities excluded from the analytical sample.

| Cities         | Province      | Geographic setting |
|----------------|---------------|--------------------|
| Anyang-si      | Gyeonggi      | Inland             |
| Bucheon-si     | Gyeonggi      | Inland             |
| Gwangmyeong-si | Gyeonggi      | Inland             |
| Pyeongtaek-si  | Gyeonggi      | Coastal            |
| Gunpo-si       | Gyeonggi      | Inland             |
| Uiwang-si      | Gyeonggi      | Inland             |
| Hanam-si       | Gyeonggi      | Inland             |
| Gimpo-si       | Gyeonggi      | Coastal            |
| Cheongdo-gun   | Gyeongsangbuk | Inland             |
| Goryeong-gun   | Gyeongsangbuk | Inland             |

**Table B.2**

First-stage city coverage by spending category, 2019.

| Category                   | Cities included |
|----------------------------|-----------------|
| Overall spending           | 152             |
| Convenience Stores         | 152             |
| Supermarkets               | 152             |
| Restaurants                | 152             |
| Food Retail                | 152             |
| Alcohol and Bars           | 152             |
| Hospitals and Pharmacies   | 152             |
| Home and Internet Shopping | 152             |
| Mixed-Exposure Recreation  | 151             |
| Indoor Recreation          | 151             |

Note: The analytical sample contains 152 cities. A city and category panel was excluded if no transactions were recorded on any day in 2019.

**Table B.3**

City-level coverage of apparent-temperature bands in 2019.

| Temperature band | Cities | Minimum | Median | P90 | Maximum |
|------------------|--------|---------|--------|-----|---------|
| <5°C             | 152    | 11      | 52     | 83  | 105     |
| 5–10°C           | 152    | 25      | 40     | 48  | 60      |
| 10–15°C          | 152    | 37      | 46     | 60  | 73      |
| 15–20°C          | 152    | 36      | 49     | 55  | 68      |
| 20–25°C          | 152    | 40      | 56     | 71  | 83      |
| 25–28°C          | 152    | 30      | 49     | 58  | 65      |
| 28–30°C          | 152    | 15      | 34     | 44  | 54      |
| 30–33°C          | 152    | 6       | 28     | 37  | 47      |
| ≥33°C            | 134    | 1       | 12     | 19  | 26      |

#### A.2. Comparison of pooled response curves using different reference bands

As shown in Fig. A.3, changing the reference temperature band re-expresses each coefficient relative to a different benchmark but leaves the fitted band relationship unchanged. Across the alternative reference intervals, the overall response curves retain the same broad nonlinear shape and warm-tail divergence between transactions and expenditure. The spline response curves provide a reference-independent representation of the same relationship.

### Appendix B. Data coverage and model diagnostics

#### B.1. Excluded cities and first-stage city coverage

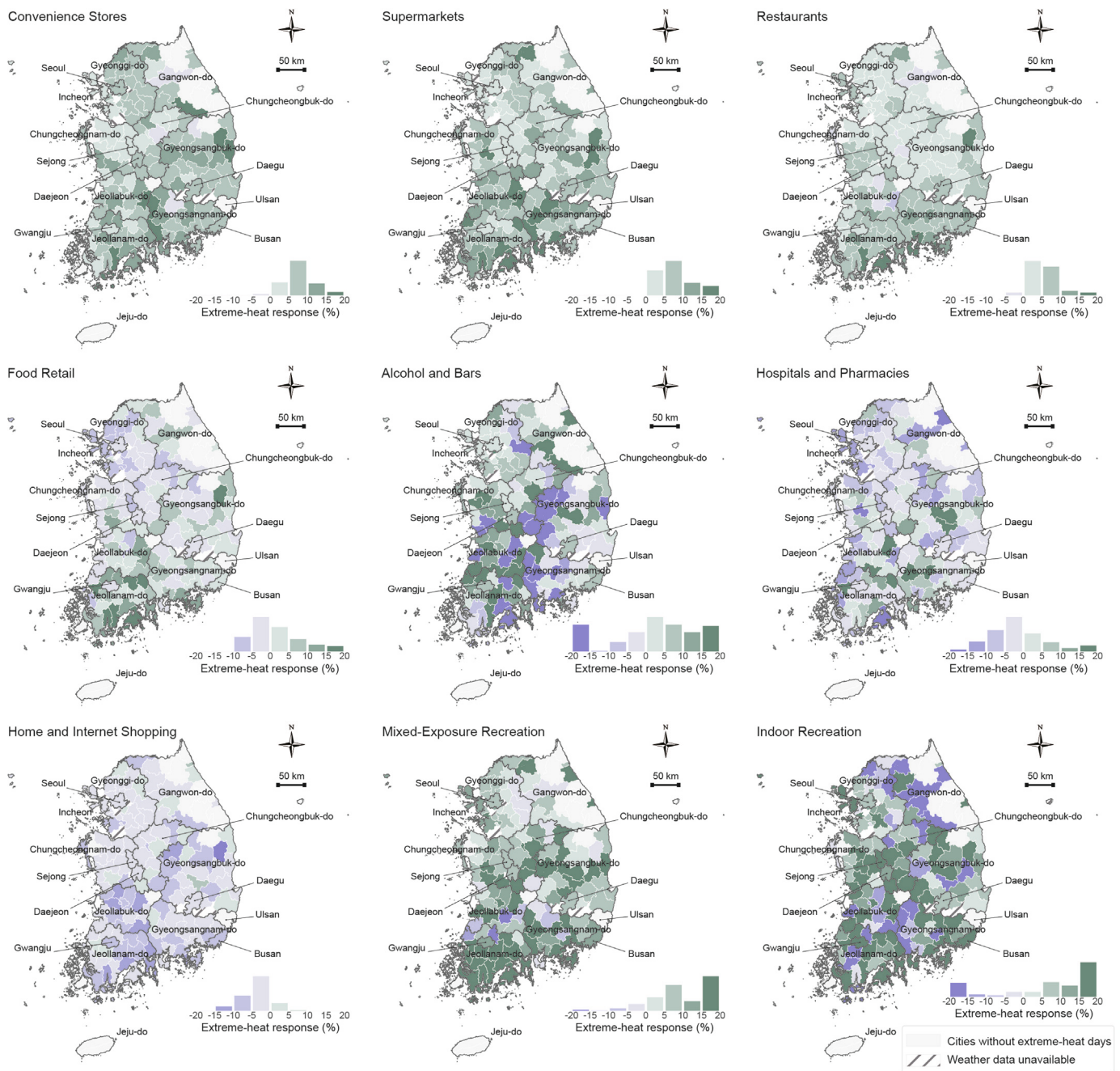
See Tables B.1 and B.2.

#### B.2. Temperature-band coverage across cities

See Table B.3.

#### B.3. First-stage model diagnostics

Because the first-stage regressions are estimated separately by city, Table B.4 summarizes diagnostics across city-specific models. Month and day-of-week controls are assessed as multi-degree-of-freedom blocks using adjusted generalized variance inflation factors (GVIFs). All other entries are ordinary VIFs.



**Fig. A.1.** Spatial distribution of category-specific transaction responses to extreme heat in South Korea, 2019. Maps report first-stage estimates for each city and category in the extreme-heat band ( $AT_{\max} \geq 33^{\circ}\text{C}$ ), expressed as percentage changes relative to the 20–25°C reference band. Each panel corresponds to one focal spending category.

#### B.4. Meta-regression covariate diagnostics

See [Tables B.5 and B.6](#).

### Appendix C. Extended-window analysis with COVID-19 controls

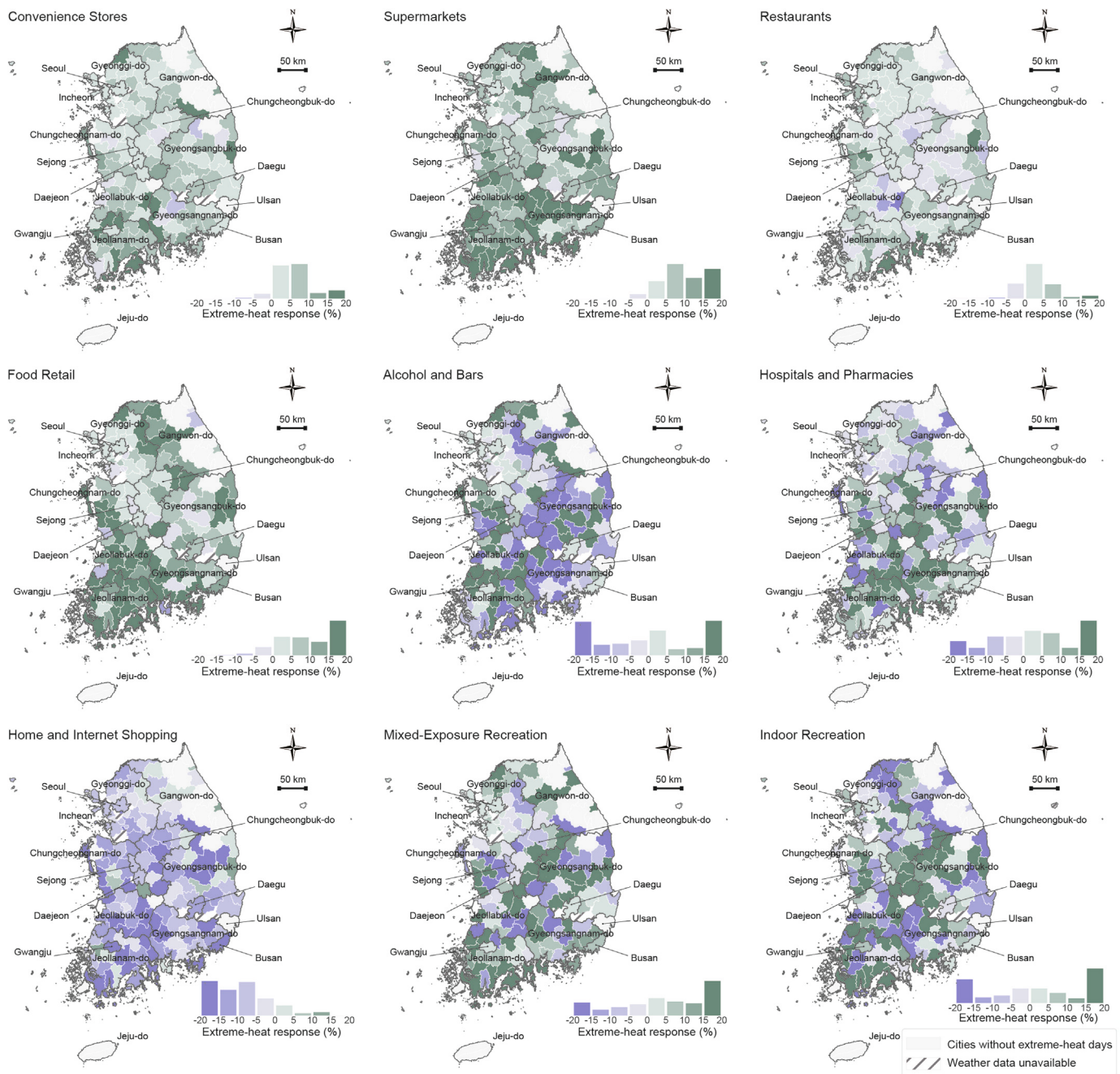
#### C.1. COVID-19 stages in the extended-window analysis

South Korea did not impose a nationwide lockdown during the 2020 study window. Its pandemic response instead combined testing, contact tracing, treatment, risk communication, and staged social distancing measures. Two major epidemic waves were separated by periods of

lower case counts. Following [Sun et al. \(2024\)](#), the extended-window models represent these changes using the five stages summarized in [Table C.1](#).

#### C.2. Temperature associations with spending in the extended-window analysis

To assess temporal robustness, the pooled response curves from the main 2019 analysis were compared with those from the extended-window analysis, which included 2020 observations and COVID-19 stage controls as described in Section 3.4. This comparison assessed whether the main relationships between weather and spending remained similar after extending the observation window to a period with



**Fig. A.2.** Spatial distribution of category-specific expenditure responses to extreme heat in South Korea, 2019. Maps report first-stage estimates for each city and category in the extreme-heat band ( $AT_{max} \geq 33^\circ\text{C}$ ), expressed as percentage changes relative to the 20–25°C reference band. Each panel corresponds to one focal spending category.

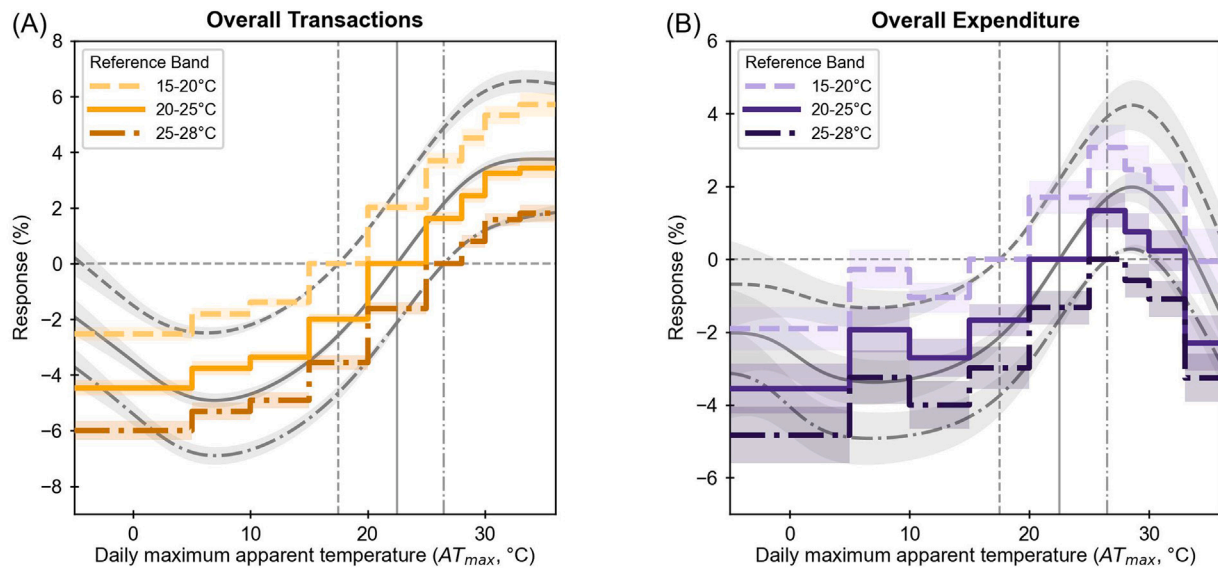
substantial disruptions to mobility, risk perception, and consumption behavior.

Fig. C.1 presents pooled response curves for overall spending. For transactions, the main and extended-window estimates showed similar overall shapes. Responses rose from the cool side toward the warm side and remained positive under extreme heat, although the extended-window warm-side increase was more modest. For expenditure, both analyses exhibited an inverted-U pattern, with lower spending at both temperature tails and higher spending near the reference band. In the main 2019 analysis, the peak occurred closer to the right boundary of the reference band, whereas in the extended-window analysis it shifted toward the cooler side. Compared with the 2019 analysis, the extended-window expenditure estimates were more negative on the cool side

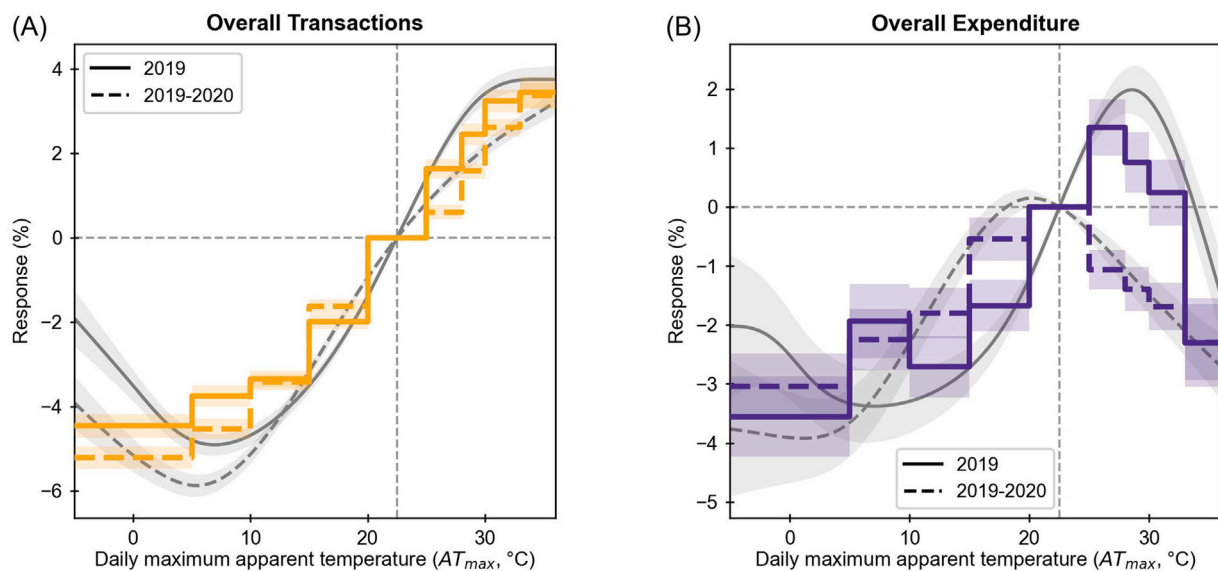
and in the heat band, while the extreme-heat estimate was nearly unchanged. Thus, the nonlinear shape remained evident, although the location and magnitude of the warm-side changes differed.

Category-specific curves in Figs. C.2 and C.3 show that the stability varied across spending categories. Essential categories, such as Convenience Stores and Supermarkets, largely maintained similar response patterns, although the magnitude of warm-side gains differed in the extended-window analysis. In contrast, categories that depended more strongly on activity setting and social participation, including Restaurants, Alcohol and Bars, and recreation categories, showed more pronounced changes in shape or amplitude.

Several deviations were particularly notable. Food Retail showed a clearer reshaping in the extended-window estimates, with transaction



**Fig. A.3.** Pooled response curves relating daily maximum apparent temperature ( $AT_{max}$ ) to overall spending in South Korea, 2019, using different reference bands. Step functions show pooled responses from the band specification, and smooth curves show pooled responses from the spline specification. Shaded areas indicate 95% confidence intervals. (A) Overall transactions. (B) Overall expenditure.



**Fig. C.1.** Comparison of pooled response curves relating daily maximum apparent temperature ( $AT_{max}$ ) to overall spending in 2019 and in the extended-window analysis (2019–2020). Step functions show pooled responses from the band specification, and smooth curves show pooled responses from the spline specification. Shaded areas indicate 95% confidence intervals. (A) Overall transactions. (B) Overall expenditure.

responses becoming more negative at both temperature tails and expenditure responses becoming broadly more negative across much of the temperature spectrum. Home and Internet Shopping showed markedly attenuated temperature sensitivity in transactions. Its extended-window expenditure response was more negative on the cool side but substantially less negative at the warm tail. Mixed-Exposure Recreation retained a warm-side increase in both spending dimensions, although its magnitude changed. Indoor Recreation retained a warm-side increase for transactions, whereas its extended-window expenditure response became negative at the warm tail.

### C.3. Precipitation and extreme-heat persistence associations in the extended-window analysis

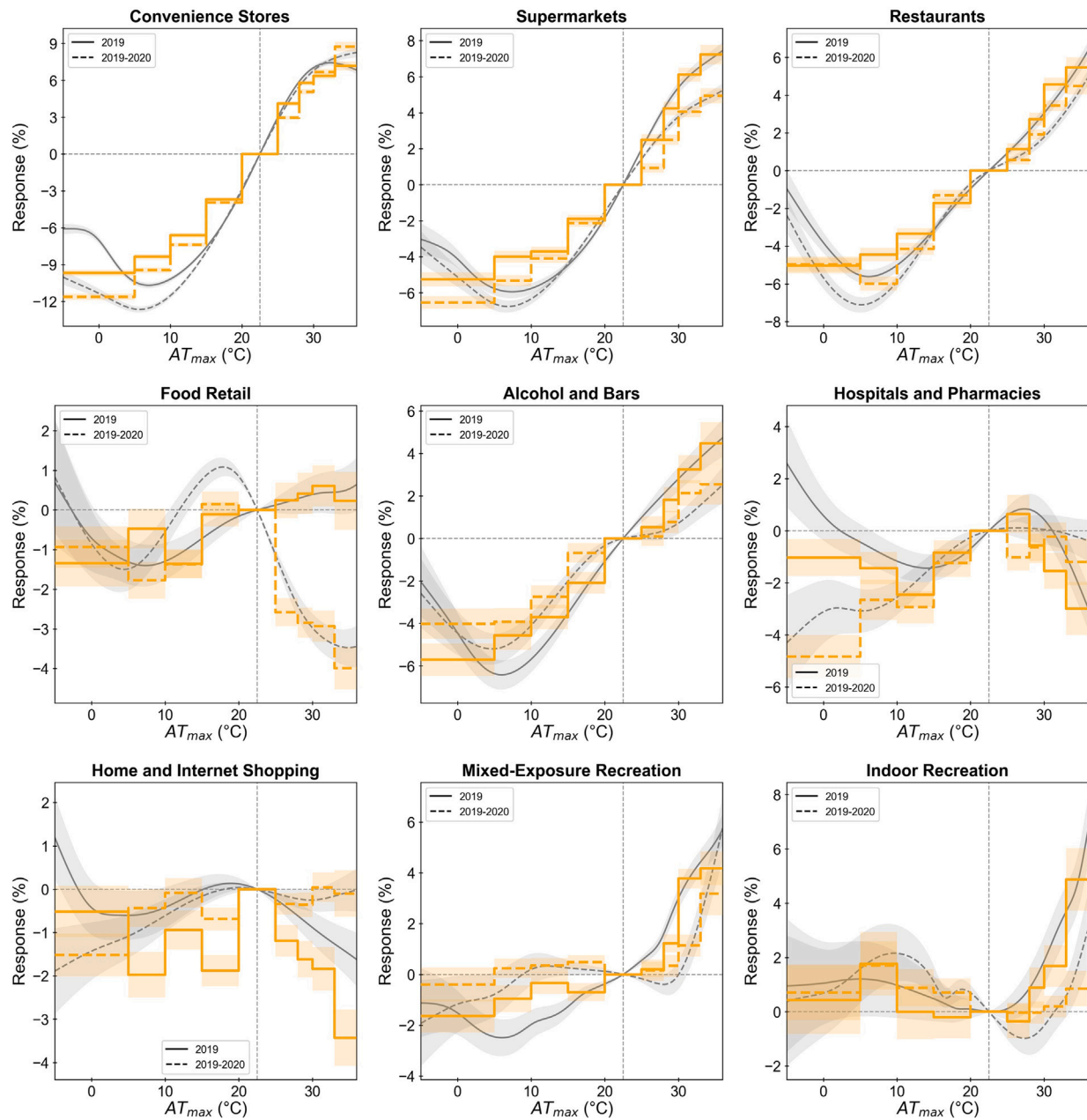
Fig. C.4 compares precipitation and extreme-heat-persistence estimates between the main and extended-window analyses. Precipitation

estimates were generally more negative in the extended-window analysis, and several positive estimates observed in 2019 became smaller. Positive extreme-heat-persistence estimates in the recreation categories also weakened. Despite these changes in magnitude, the differences across spending categories remained evident.

## Appendix D. Temporal displacement around extreme-heat events

### D.1. Analytical design

An extreme-heat event is defined as one or more consecutive days with  $AT_{max} \geq 33^\circ\text{C}$ . The term event therefore includes both single-day and multi-day occurrences. The two tests examine complementary forms of short-term temporal displacement around these events. The first re-estimates the main models after excluding adjacent days without

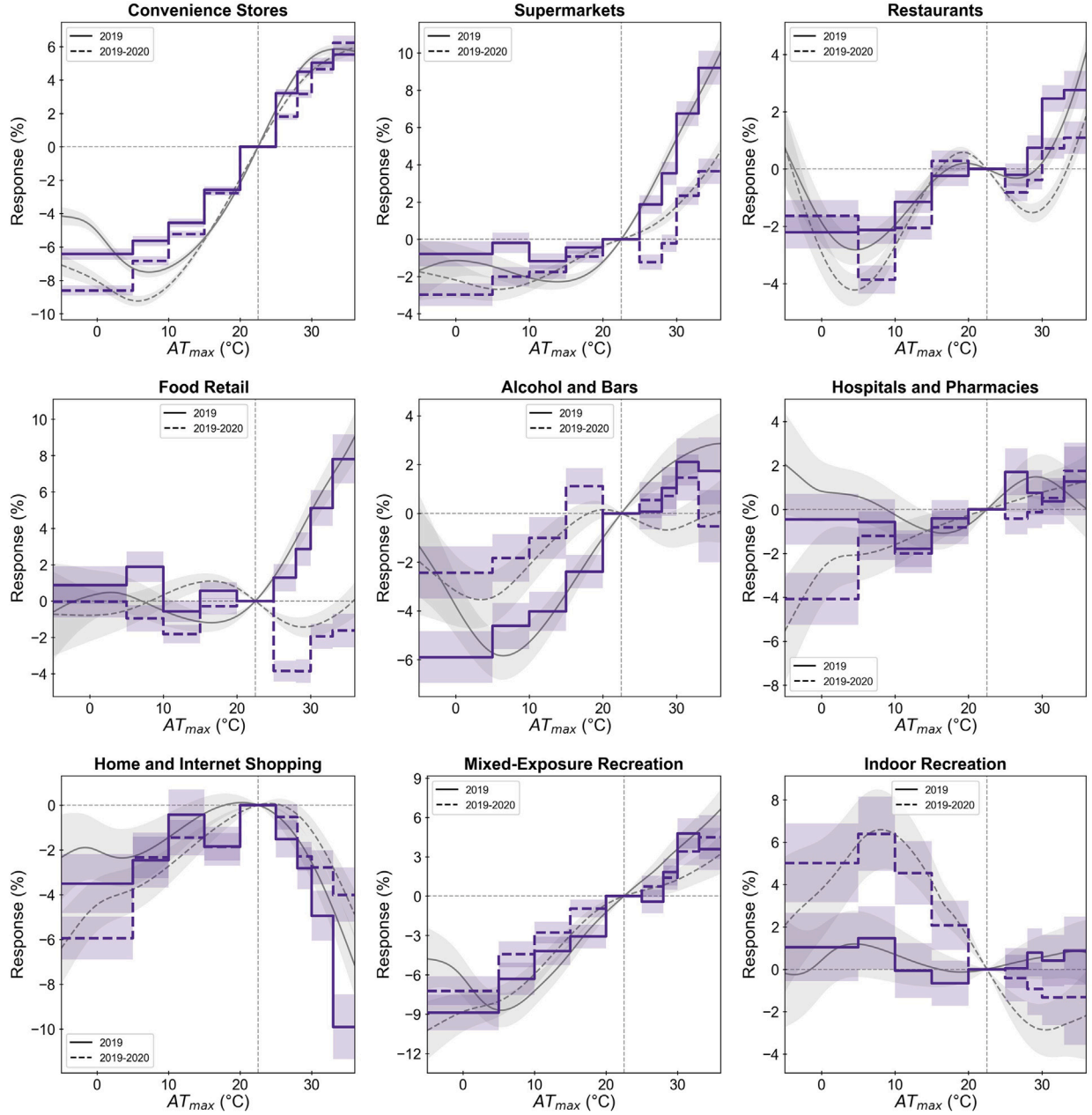


**Fig. C.2.** Category-specific comparison of pooled response curves for transactions in the main 2019 analysis and the extended-window analysis (2019–2020). Each panel corresponds to one focal spending category. Responses are expressed as percentage changes relative to the 20–25°C reference band. Step functions show pooled responses from the band specification, smooth curves show pooled responses from the spline specification, and shaded areas indicate 95% confidence intervals.

extreme heat to assess whether their inclusion influences the extreme-heat estimates in the main analysis. The second examines whether spending on adjacent days shows systematic pre-event accumulation or post-event rebound.

The first test re-estimates the city-specific models after removing adjacent days without extreme heat within three calendar days before event onset or after event end for all 559 events identified in 2019. Extreme-heat days are retained, and overlapping exclusion windows are combined. The models otherwise follow the main specification and pooling procedure. They are re-estimated for all 152 cities, with pooled extreme-heat estimates based on cities in which the  $\geq 33^\circ\text{C}$  band is observed.

The second test estimates an event profile for overall spending around isolated events. Days  $-3$  to  $-1$  are anchored to event onset, while days  $+1$  to  $+3$  are anchored to event end so that days within a multi-day event are not classified as post-event adjustment. All six days surrounding an event must be days without extreme heat, the expanded windows must not overlap, and all event days are excluded from the event regression. Dates surrounding events that do not meet these restrictions are also excluded from the comparison group. This procedure retains 214 isolated events across 119 cities. The reference group comprises other days without extreme heat in the same cities during May–September 2019 that fall outside every excluded event window. The three-day horizon focuses on immediate adjustment in purchase



**Fig. C.3.** Category-specific comparison of pooled response curves for expenditure in the main 2019 analysis and the extended-window analysis (2019–2020). Each panel corresponds to one focal spending category. Responses are expressed as percentage changes relative to the 20–25°C reference band. Step functions show pooled responses from the band specification, smooth curves show pooled responses from the spline specification, and shaded areas indicate 95% confidence intervals.

timing, limits overlap between nearby events, and corresponds to the exposure horizon used to construct  $EHP_{ct}$ .

$$\ln(1 + Y_{ct}) = \mu_{c \times m(t)} + \tau_t + f(AT_{max,ct}) + \delta EHP_{ct} + \gamma P_{ct} + \phi W S_{ct} + \sum_{r=-3}^{-1} \pi_r Pre_{ct}^r + \sum_{r=1}^3 \lambda_r Post_{ct}^r + \epsilon_{ct}. \quad (8)$$

Here,  $\mu_{c \times m(t)}$  denotes city-specific month fixed effects and  $\tau_t$  denotes date fixed effects.  $f(AT_{max,ct})$  is a natural cubic spline with the same internal knots as the main spline specification, and  $EHP_{ct}$  controls for extreme-heat exposure during the preceding three days.  $Pre_{ct}^r$  and  $Post_{ct}^r$  identify event position relative to event onset and end, respectively. The model is estimated separately for overall transactions and expenditure, with standard errors clustered by city and date. In addition to the day-specific coefficients, we test the average log-scale differences

over the three-day pre-event and post-event windows, defined as  $\bar{S}_{pre} = 3^{-1} \sum_{r=-3}^{-1} \pi_r$  and  $\bar{S}_{post} = 3^{-1} \sum_{r=1}^3 \lambda_r$ .

## D.2. Results

**Fig. D.1** compares the extreme-heat estimates from the main analysis with those obtained from models excluding adjacent days without extreme heat. For overall transactions, the pooled estimate was nearly unchanged, increasing from 3.44% to 3.58%. For overall expenditure, the estimate remained negative and statistically significant, while its magnitude increased from −2.30% to −3.25%. The estimated direction was retained in most comparisons across categories and spending dimensions. The only sign change occurred for Alcohol and Bars expenditure, which changed from 1.73% to −0.53%

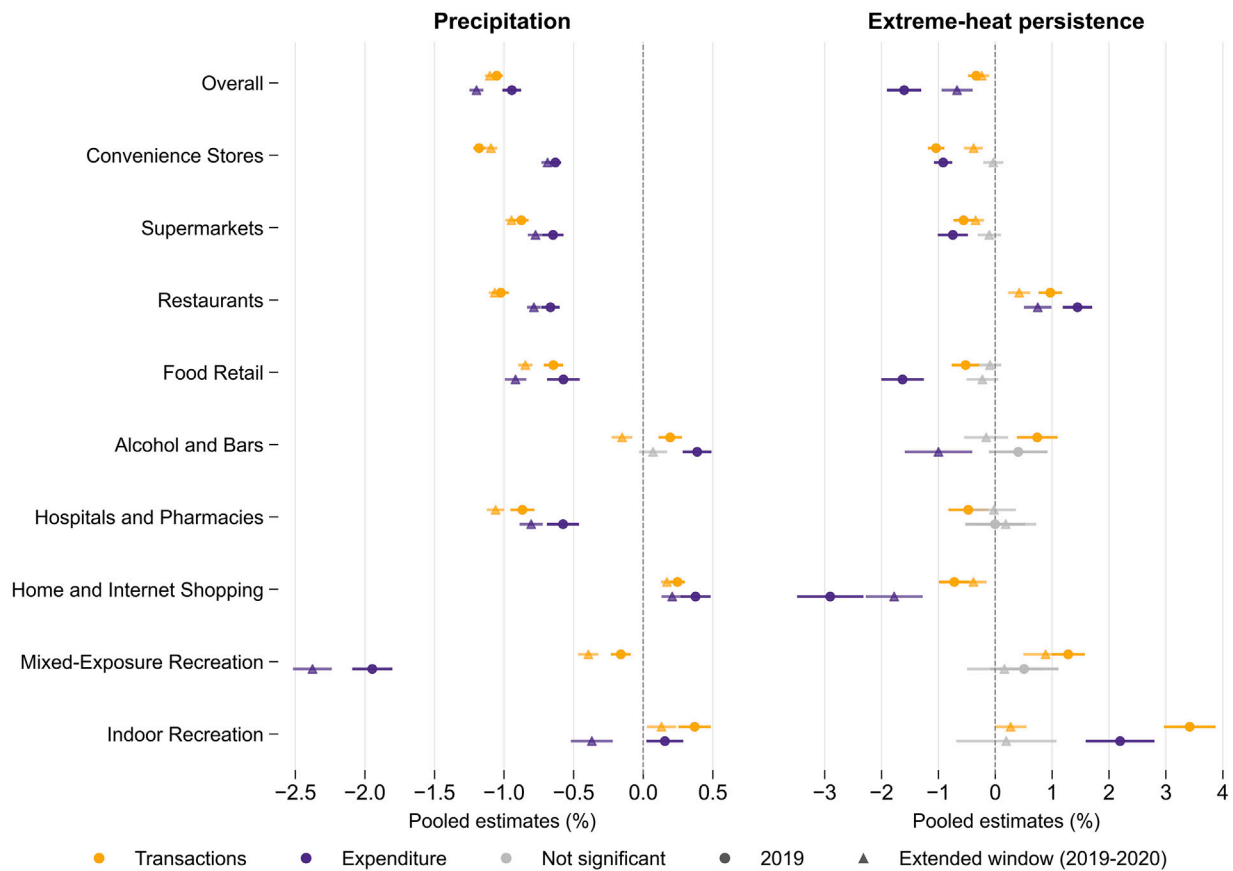


Fig. C.4. Category-specific pooled estimates for precipitation ( $P_{ct}$ ) and extreme-heat persistence ( $EHP_{ct}$ ) in the main 2019 analysis and the extended-window analysis. Points denote pooled estimates, and horizontal bars denote 95% confidence intervals.

Table B.4

Collinearity diagnostics for city-specific first-stage models.

| Variable                                | Diagnostic    | Cities | Median | P90  |
|-----------------------------------------|---------------|--------|--------|------|
| Month fixed effects                     | Adjusted GVIF | 152    | 1.22   | 1.24 |
| Day-of-week fixed effects               | Adjusted GVIF | 152    | 1.01   | 1.02 |
| Holiday indicator                       | VIF           | 152    | 1.07   | 1.09 |
| Precipitation ( $P_{ct}$ )              | VIF           | 152    | 1.20   | 1.31 |
| Extreme-heat persistence ( $EHP_{ct}$ ) | VIF           | 134    | 2.02   | 3.28 |
| Wind speed ( $WS_{ct}$ )                | VIF           | 152    | 1.35   | 1.58 |
| Temperature band: <5°C                  | VIF           | 152    | 6.73   | 9.65 |
| Temperature band: 5–10°C                | VIF           | 152    | 4.59   | 5.58 |
| Temperature band: 10–15°C               | VIF           | 152    | 4.04   | 4.90 |
| Temperature band: 15–20°C               | VIF           | 152    | 2.69   | 3.25 |
| Temperature band: 25–28°C               | VIF           | 152    | 2.32   | 2.66 |
| Temperature band: 28–30°C               | VIF           | 152    | 2.45   | 2.89 |
| Temperature band: 30–33°C               | VIF           | 152    | 2.57   | 3.19 |
| Temperature band: ≥33°C                 | VIF           | 134    | 2.38   | 3.63 |

and became statistically indistinguishable from zero. Although some confidence intervals widened after observations were removed, the principal category patterns were preserved. The pooled  $EHP_{ct}$  estimates also remained negative overall, changing from  $-0.34\%$  to  $-0.46\%$  for transactions and from  $-1.60\%$  to  $-0.94\%$  for expenditure.

Fig. D.2 reports the overall event profile after conditioning on realized temperature and recent extreme-heat exposure. Transactions and expenditure were lower on day  $-3$  by  $1.39\%$  and  $2.11\%$ , respectively, while estimates on days  $-2$  and  $-1$  were close to zero and not statistically significant. The average three-day pre-event contrasts were also not statistically significant for transactions ( $\bar{S}_{pre} = -0.42\%$ , 95% CI:  $-1.36\%$  to  $0.52\%$ ,  $p = 0.379$ ) and expenditure ( $\bar{S}_{pre} = -1.24\%$ , 95% CI:  $-2.74\%$  to  $0.27\%$ ,  $p = 0.107$ ). This pattern does not indicate positive spending accumulation immediately before event onset.

The estimates on day  $+1$  did not reach the 5% significance level and were followed by estimates that were also not statistically significant on days  $+2$  and  $+3$ . The average three-day post-event contrasts were likewise small and imprecisely estimated for transactions ( $\bar{S}_{post} = 0.25\%$ , 95% CI:  $-1.49\%$  to  $2.02\%$ ,  $p = 0.780$ ) and expenditure ( $\bar{S}_{post} = 0.37\%$ , 95% CI:  $-2.72\%$  to  $3.56\%$ ,  $p = 0.816$ ).

### D.3. Interpretation

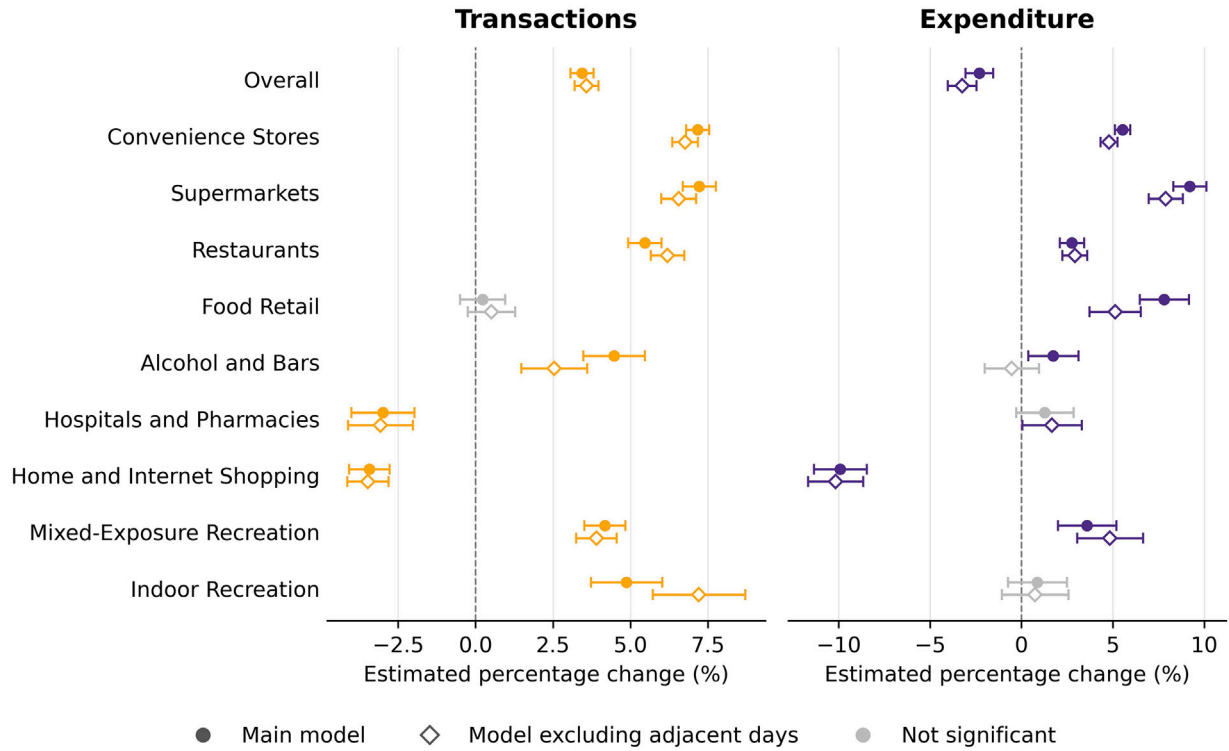
Excluding adjacent days without extreme heat preserved the main pattern of higher transaction frequency and lower expenditure under extreme heat. This result indicates that the extreme-heat estimates in the main analysis are unlikely to be driven primarily by adjacent days that may have captured spending shifted from extreme-heat days.

The isolated-event profile provides no evidence of positive spending accumulation immediately before event onset. Although transactions and expenditure were lower on day  $-3$ , the estimates on days  $-2$  and  $-1$  were close to zero, and the average pre-event contrasts were not statistically significant. The post-event estimates similarly provide no evidence of a systematic three-day rebound. The positive but imprecise day- $+1$  coefficients may be consistent with a brief adjustment rather than sustained compensation.

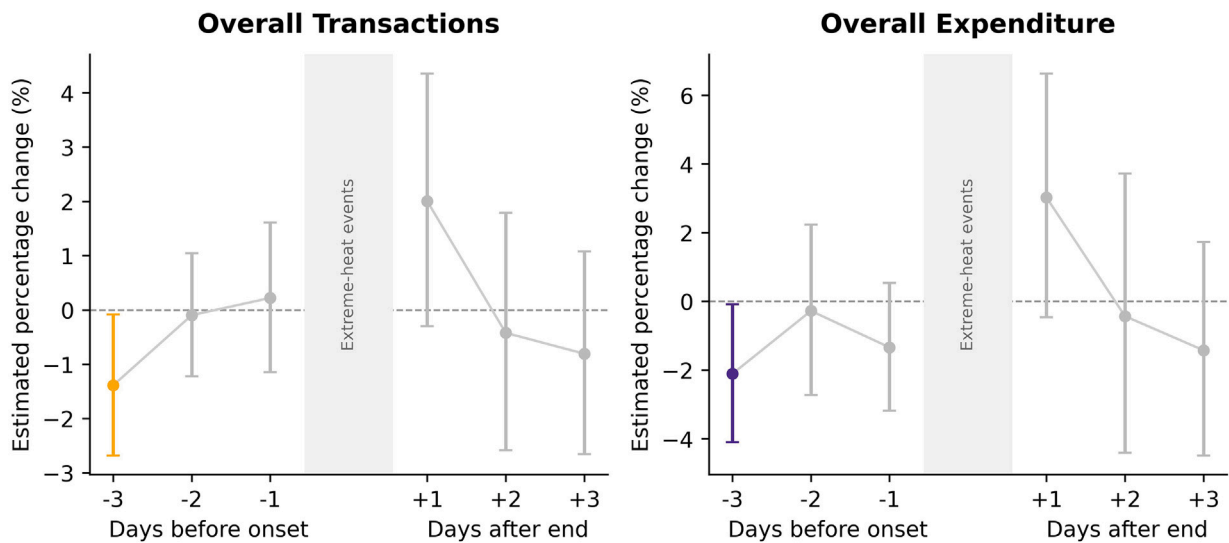
Because the dataset does not contain the weather forecasts available to consumers, the pre-event coefficients cannot be interpreted as direct evidence of forecast-driven anticipation. In addition, the event-profile analysis is restricted to isolated events with complete and non-overlapping windows, whereas the models excluding adjacent days use all identified events. Taken together, the two analyses do not indicate systematic pre-event accumulation or post-event rebound within the three-day windows examined here. Short-term temporal displacement is therefore unlikely to be a primary explanation for the extreme-heat estimates in the main analysis.

**Table B.5**  
Descriptive statistics for city attributes used in the meta-regressions.

| Attribute                                     | Mean   | SD      | Min   | P25   | Median | P75    | Max       |
|-----------------------------------------------|--------|---------|-------|-------|--------|--------|-----------|
| Coastal                                       | 0.35   | 0.48    | 0.00  | 0.00  | 0.00   | 1.00   | 1.00      |
| ClimTemp (°C)                                 | 12.56  | 1.30    | 8.63  | 11.68 | 12.68  | 13.38  | 16.20     |
| Share65 (%)                                   | 23.25  | 8.47    | 8.52  | 15.66 | 23.61  | 29.97  | 39.94     |
| GDP per capita (million KRW)                  | 38.12  | 24.98   | 12.21 | 26.16 | 33.43  | 40.77  | 208.59    |
| Population density (persons/km <sup>2</sup> ) | 802.08 | 1911.69 | 19.60 | 66.02 | 144.72 | 570.08 | 16 062.77 |



**Fig. D.1.** Comparison of pooled extreme-heat estimates from the main models and models excluding days without extreme heat within three calendar days before event onset or after event end. Circles show estimates from the main models, diamonds show estimates after adjacent days were excluded, and horizontal bars indicate 95% confidence intervals. Colored estimates are statistically significant at the 5% level, while gray estimates are not statistically significant.



**Fig. D.2.** Estimated changes in overall transactions and expenditure before the onset and after the end of isolated extreme-heat events. Extreme-heat days are excluded, and the shaded center represents the excluded event interval separating the window anchored to event onset from the window anchored to event end. Colored points indicate estimates statistically significant at the 5% level, while gray points indicate estimates that are not statistically significant. Vertical bars show 95% confidence intervals.

Table B.6

Pairwise correlations and variance inflation factors for city attributes used in the meta-regressions.

| Attribute | Coastal | ClimTemp | Share65 | lnGDPpc | lnPopDens | VIF   |
|-----------|---------|----------|---------|---------|-----------|-------|
| Coastal   | 1.000   | 0.503    | 0.028   | 0.128   | 0.052     | 1.391 |
| ClimTemp  | 0.503   | 1.000    | −0.099  | −0.003  | 0.298     | 1.566 |
| Share65   | 0.028   | −0.099   | 1.000   | −0.185  | −0.809    | 3.503 |
| lnGDPpc   | 0.128   | −0.003   | −0.185  | 1.000   | −0.009    | 1.157 |
| lnPopDens | 0.052   | 0.298    | −0.809  | −0.009  | 1.000     | 3.672 |

Table C.1

COVID-19 stages in South Korea used to define the stage control in the extended-window analysis.

| Stage | Label                | Timing in 2020 | Brief description                                                                                                                                                                                           |
|-------|----------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | Normal period        | Jan 1–Feb 17   | Initial stage of the pandemic, when confirmed cases were mainly limited to overseas travelers or their close contacts.                                                                                      |
| 1     | First wave           | Feb 18–May 5   | Marked by a sharp increase in community transmission, especially in Daegu and Gyeongsangbuk-do, together with intensified social-distancing guidance.                                                       |
| 2     | First stable period  | May 6–Aug 13   | Lower case counts, with small-scale community infections mainly originating in the Seoul metropolitan area. “Distancing in Daily Life” began, and a new three-level social-distancing scheme was announced. |
| 3     | Second wave          | Aug 14–Sep 20  | Triggered by multiple hotspots in the Seoul metropolitan area and followed by nationwide spread. Level 2 social distancing was introduced on Aug 23.                                                        |
| 4     | Second stable period | Sep 21–Nov 1   | Stabilization following the second wave. New cases declined, although level 2 social distancing remained in place until Oct 11 before being relaxed.                                                        |

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